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Flight to cryptos: Evidence on the use of cryptocurrencies in times of geopolitical tensions

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ABSTRACT

This paper examines trading activity in cryptocurrencies in times of geopolitical crises. Cryptocurrencies represent speculative assets as well as payment methods. This combination of features is not present in securities like stocks and bonds. The empirical analysis is based on a sample of 93 events associated to potential limitations of a fiat currency circulation and considers five cryptos (Bitcoin, Ether, Ripple, Dash, and Tether). We find that trading in cryptocurrencies increases with events of geopolitical tensions. The increase in cryptocurrencies trading in times of crises can be motivated by different explanations (e.g., protecting savings as the domestic currency devaluates, making payments as the domestic financial system is no longer available, avoiding sanctions) which are difficult to disentangle. A more specific analysis concerning the EU sanctions established in 2022 on Russia shows that crypto trading slows down when crypto-related services (wallet, account or custody services) are explicitly included in EU financial sanctions packages. A warning about data limitations: the data set only includes trading activities conducted on centralized exchanges (CEXs) and does not include transactions conducted on decentralized exchanges (DEXs). Sanctions are supposed to be more effective on CEXs where the platform acts as a custodian for trader's asset. We also examine trading activity from the Ukranian Hrvynia to cryptocurrencies and find a strong increase in outflow from Hrvynia since the beginning of the conflict. This finding – which is not affected by donations of cryptos received by Ukraine from abroad – is consistent with the hypothesis that Ukrainians increasingly exchanged their domestic currency for cryptocurrencies.

1. Introduction

Cryptocurrencies are now part of the financial system, whether as a form of speculative asset or as a payment tool, they prove to respond to investors' needs (Auer et al., 2022). For example, Bitcoin's relatively high returns and its low correlation with popular assets like global equities or bonds, both in regular times and periods of financial distress, make it an excellent diversification instrument (Baur, Hong, & Lee, 2018).

The evidence that cryptocurrencies are here to stay is proved by the significant shift of trending topics about cryptos we have witnessed in the last years. Initially, the issue was whether cryptocurrencies were a scam ready to deflate and/or a tool only interesting

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for financial crimes and money laundering activities. Nowadays, the central topic has shifted towards a more realistic and productive approach, i.e. the kind of regulation that we want to adopt for cryptocurrencies.[1]

In this paper we study a specific use for cryptocurrencies, whether they represent an alternative payment method when the circulation of legal tender currencies is threatened or limited due to external shocks such as international sanctions or military invasion. For example, with US dollar representing more than 40 percent of global payments as of December 2021 (ECB, 2022), limiting the access to US dollar currency results in an actual freeze of the assets for the individuals undergoing the sanction. An alternative currency that provides fast and disintermediated exchange of funds could offer a solution for the individuals to evade sanctions.

The case for cryptocurrencies as the future of international payments traces back to Bitcoin's origins (Nakamoto, 2008), as well as relates with most recent cryptocurrencies with the ambition to replace SWIFT as the next system of international payments (Smith, 2022), although not in the short-term (Qiu, Zhang, & Gao, 2019). However, a widespread diffusion of cryptocurrency, especially in the form of stablecoins, might involuntarily reshape the geopolitical activity, produce its effect on a strategic and military field (Fantacci & Gobbi, 2020), threaten the stability of the financial system and hamper monetary sovereignty (Panetta, 2020).

Recent developments provide a useful example. Following the February 2022 Russian invasion in Ukraine, severe sanctions were implemented to restrict Russian citizens' ability to move their cash holdings, consequently the exchange of Russian Ruble to Bitcoins suddenly increased, with the latter rapidly nicknamed "conflict currency" (Singh & Mattackal, 2022). The magnitude of the trades was so significant that regulating authorities as well as crypto asset trading platforms proposed the extension of the sanctions to cryptocurrencies to avoid loopholes in the approved sanctions and regulators undertook the challenge on both sides of the Atlantic (Ostroff, 2022). In Europe, the European Union agreed to an extra package of restrictive measures against Russia with the prohibition to provide high-value crypto-asset services to Russia (EU, 2022a), furtherly strengthen by the end of 2022 (EU, 2022b). In the United States, several authorities acted in the same direction.[2]

To understand the magnitude of the above phenomenon, Fig. 1 presents daily trading activity from Ruble to Bitcoin and other four cryptocurrencies (Ether, Ripple, Dash, and Tether) from early 2022 to the end of February 2022, and Fig. 2 shows the overall volume in millions USD exchanged from Ruble to all the before mentioned cryptocurrencies since the start of the war to the end of 2022. Starting from 2014 the Russian-Ukrainian territorial dispute offers several single day events where sanctions on Russian citizens were either applied or expected. Based on this time series and the strong reliance of Russian economy on foreign currency reserves, either in individual accounts or in the Central Bank of Russian Federation (CBR) account, our sample is sufficient to study the behavior of individuals threatened with sanctions limiting their ability to exchange domestic currency for a foreign currency.

The aim of this paper is to investigate whether Bitcoin and other relevant cryptocurrencies were used to mitigate the effects of financial sanctions. Our contribution is important for policy makers since it addresses the issue of whether cryptocurrencies need a stronger regulatory framework for Anti-Money Laundering (AML) and Countering the Financing of Terrorism (CFT) purposes, as well as whether financial sanctions in force are able to stop the exchange of fiat currency for cryptos. We are not aware of other studies investigating the above research question. This lack of research may be explained by the novelty of the topic and the strong link to current events.

As for our results, we find a sharp increase of the outflow of Ruble to cryptocurrencies as individuals, either in response to or in anticipation of sanctions, wanted to mitigate their negative effects. Liquidity plays a crucial role in the above setting, since our evidence suggests that Bitcoin and Ether attract most of the traders. On the other hand, additional features associated to other cryptocurrencies, either positive or negative, play a minor role in the decision to exit from a currency subject to financial sanctions. We also notice a reduction in trades after the first application of financial sanctions on crypto assets; however, it is hard to disentangle whether the measure was effective or if it was like closing the stable door after the horse has bolted.

The remainder of this paper is organized as follows: Section 2 describes the sample, presents the research design, and states the testable hypotheses, Section 4 presents the results of our empirical analysis, and finally Section 5 concludes.

2. Sample and data

2.1. Sample selection

Our sample includes news about events that resulted, or potentially resulted, in a severe limitation of currency circulation for the

[1] The range of approaches spans from countries such as China that prohibits any kind of use for these currencies to countries like El Salvador ready to adopt Bitcoin as legal tender, and US in the middle providing, what it seems, a soft regulation to let the business grow, but highlighting the risk for evasions via the use of digital assets (The White House, 2022).

[2] Warren et al. (2022) wrote a letter to Secretary Yellen inquiring about the Department of the Treasury's progress enforcing sanctions compliance by the cryptocurrency industry and calling for a stronger enforcement of procedures by the Treasury's Office of Foreign Assets Control (OFAC). DOJ (2022) announced the launch of a task force dedicated to enforcing sanctions and restrictions placed in response to Russia's actions in Ukraine, specifically targeting efforts to use cryptocurrency to evade US sanctions. OFAC (2022) issued a FAQ to reaffirm that compliance on Russian sanctions is required regardless of whether a transaction is denominated in traditional fiat currency or virtual currency and that the broad enforcement authorities will be used to act against any violations of the guidance for the virtual currency industry released by OFAC (2021). Still within the perimeter of the Department of the Treasury, Financial Crimes Enforcement Network (FinCEN) issued earlier on an alert to financial institutions providing examples of red flags that indicate suspected sanctions evasions, with three of the red flags directly related to the use of cryptocurrency (FinCEN, 2019), and more recently advised an increased vigilance from financial institution for potential Russian sanctions evasion attempts (FinCEN, 2022).

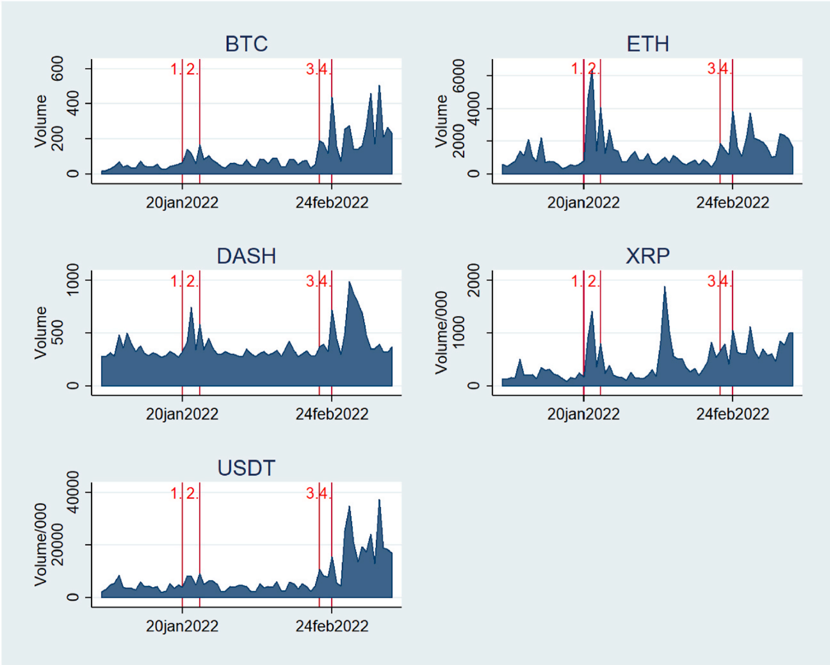

Fig. 1. Volume from Ruble to destination cryptocurrency in the eve of Ukraine invasion
The Figure presents volume in destination cryptocurrency for trades exchanging Russian Ruble for the Bitcoin (BTC), Ether (ETH), Dash (DASH), Ripple (XRP) and Tether (USDT) between early January 2022 and mid-March 2022. Within the represented timeframe we evidence four different relevant events: 1. On January 20, 2022 The White House releases a statement alerting that Russia might invade Ukraine (https://www.reuters.com/world/europe/biden-says-any-russian-movement-into-ukraine-will-be-considered-invasion-2022-01-20/), 2. on January 24, 2022 NATO sends reinforcements to eastern Europe and U.S. puts troops on military alert (https://www.reuters.com/world/europe/nato-sends-ships-jets-eastern-europe-ukraine-crisis-2022-01-24), 3. on February 21, 2022 Russian President Vladimir Putin announced that Russia recognizes the independence of two pro-Russian breakaway regions in eastern Ukraine (https://www.reuters.com/world/europe/putin-plans-recognise-eastern-ukraines-breakaway-regions-kremlin-2022-02-21/) and 4. on February 24, 2022 Russian 'special military operation' against Ukraine is authorized (https://www.reuters.com/world/europe/russias-putin-authorises-military-operations-donbass-domestic-media-2022-02-24/).

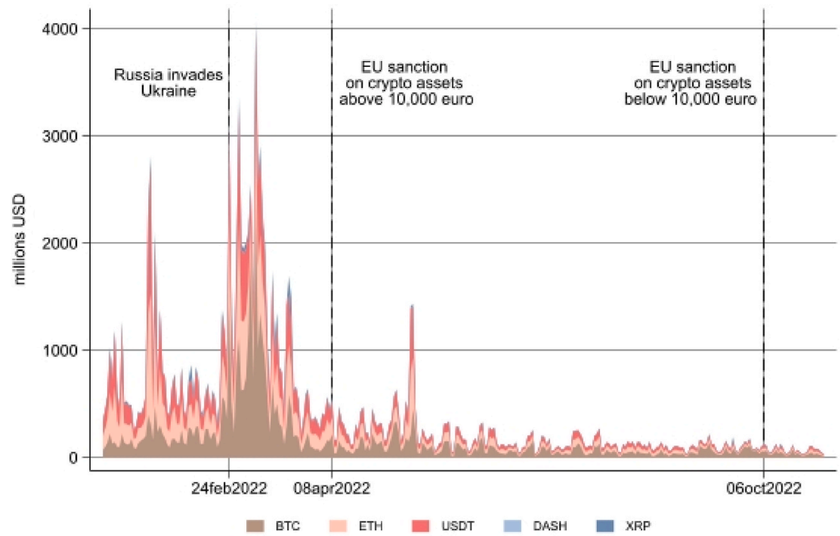

Fig. 2. Overall volume from Ruble to destination cryptocurrency since the start of the war
The Figure presents volume in millions USD for trades exchanging Russian Ruble for the Bitcoin (BTC), Ether (ETH), Dash (DASH), Ripple (XRP) and Tether (USDT) between early January 2022 and mid-November 2022. The Figure highlights the three most significant events for our analysis, the invasion of Ukraine on February 24, 2022, the first application of sanctions to Russia regarding crypto assets by EU on April 8, 2022, and the extension of the same sanctions on October 6, 2022.

targeted country (i.e., the country concerned with the sanctions) and a stop to the exchange of the domestic currency with other fiat currencies.

Events selection is carefully conducted and verified via alternative databases as follows. We first build our database from Refinitiv 'NEWS' App scanning for news related to the exchange rate between US dollar and Russian Ruble (topic "RUB = ") from January 1st, 2013 to November 1st, 2022.

To have a significant number of news related to events of geopolitical tensions, the sample period covers the conflict between Russia and Ukraine. The beginning of the sample (i.e., January 1st, 2013) is more than one year before the first significant news related to the tensions between Russia and Ukraine, whose early signals date back to February 2014. The end of the sample (i.e., November 1st, 2022) is such that the sample covers the start of the war on February 24, 2022 as well as the crypto-related sanctions established on April 8 and October 6, 2022.

We narrow the perimeter by selecting information associated with specific topics such as Embargoes and Sanctions (topic "TRDEMB"), Military conflict ("WAR"), Ukraine ("UA"), Bombing ("BOMB") and Civil Unrests ("VIO").[3] From this initial gathering of news we hand-collected news related to direct communication of international sanctions applications, and news related to targeted country demeanor that might result in a short term application of international sanctions, such as civil unrest in disputed territories and the consequential repression by the targeted country and/or by the threatened country, military drills conducted by the targeted country close to disputed territories, and proven military/unproven non-military incidents involving targeted country. Only the first appearing news is selected, while any follow-up articles are excluded.

The final sample is composed by 93 events. Table 1 provides a short description for each event and reports the frequency of each category of events that make our sample. The sample is mainly populated by news of the application of sanctions by foreign countries and supernational institutions to individuals of the targeted country (62% of the total events). Table A. 1 in Appendix provides a detailed description for each of the 93 events in our sample. Given the high number of sanctions news in our sample, we double check the occurrence of these events using an alternative database provided by OpenSanctions.org.[4]

Table 2 provides a description of the most relevant restrictive measures adopted by EU against Russia in the period 2014–2022. The EU sanctions mainly concern assets freeze and travel bans during 2014 and more draconian ban on economic and financing activities from 2022. The second most relevant category in our sample concerns military drills conducted by the targeted country and allies close to disputed territories, single military fight and starting of military operation (20% of the total events). Whereas for news on sanctions application the link to our research topic is direct, with military drills and other non-sanctions event we need to provide an additional justification for their inclusion. Military exercises, Russian in particular, are far from a yearly routine event performed just to stretch soldiers' legs, they are more like a dress rehearsal for an upcoming military invasion, led or suffered (Depledge, 2020). When performed by Russia these events are carefully followed by NATO and perceived as a potential threat that increases instability and contribute to an escalation that might grow into an invasion (see Clem, 2018; Hughes, 2021). On the other hand, NATO itself relies on military war games as a deterrence strategy towards Russia (see Depledge, 2020; Kubai, 2022). In this framework and given the active alert related to the Russian-Ukrainian territorial dispute ongoing since 2014, any form of military exercises might be perceived by targeted individuals as a precursor to active military operation and the consequential implementation of international sanctions on targeted country. We build the main variable of our study based on this list of events; the variable takes a value of one for every event day.

2.2. Data description

We collect daily volume of trades from January 1st, 2013 to November 1st, 2022. This trading activity represents flows from the currency subject to international sanctions that limit the free exchange of funds (i.e., the Russian Ruble (RUB)) to a panel of five relevant cryptocurrencies (i.e., Bitcoin (BTC), Ether (ETH), Ripple (XRP), Dash (DASH) and Tether (USDT)). We focus on more than one cryptocurrency to provide robust results, avoiding idiosyncratic noise from the cryptocurrencies and investigate whether our hypothesis is blockchain agnostic or individuals have preference for one crypto or the other.

We now report the motivations behind the selection process of the five cryptocurrencies. We select Bitcoin and Ether as the two most significant cryptocurrencies by market capitalization as of November 2022, according to Coinmarketcap.com. Bitcoin appears to be a cryptocurrency where liquidity is mostly driven by its own network and is only weakly influenced by US state of the economy (Scharnowski, 2021). Bitcoin is perceived as complementary to the assets of gold and presents hedging capabilities like it. It can be used against the stock exchange index and the US dollar in the short term (Dyhrberg, 2016a, 2016b), although with higher volatility (Baur, Hong, & Lee, 2018; Baur & Hoang, 2021). Bitcoin can be used as measure to hedge against policy uncertainty and can be useful during extreme bearish and bullish markets (We et al., 2019). Yet, the cryptocurrency hardly qualifies as a better solution than gold when investors search for safety, being more volatile, less liquid, and more expensive to trade (Smales, 2019).

As for Ether, the cryptocurrency is representative of the smart contract-oriented economics. Additionally, we select Ripple, a crypto currency that runs on a semi-permissioned blockchain, as representative of a banking-oriented crypto payment system[5] and Dash as representative of privacy-oriented coins, where users can opt to make a transaction anonymous (Duffield & Diaz, 2016; Duffield &

[3] The command line we provide in Refinitiv browser is the following: R:RUB = AND (Topic:TRDEMB OR Topic:WAR OR Topic:UA OR Topic: BOMB OR Topic:VIO).

[4] The database is available at: https://www.opensanctions.org/datasets/sanctions/.

[5] See: "What is XRP (Ripple)?", Forbes Advisor, May 31, 2022. https://www.forbes.com/advisor/investing/cryptocurrency/what-is-ripple-xrp/.

Table 1
Events concerning currency limitations in Russia The Table reports the breakdown by categories of the 93 relevant events concerning currency limitation in Russia from 2013 to early 2022.

Categories	Description	N. Observations
Central banking intervention	Intervention by the targeted country's central bank to limit the effect of devaluation on domestic currency due to economic fragility linked to probable upcoming military operations and/or sanctions application by foreign countries and supernational institutions.	1
Civil unrest	Civil protests in targeted country amid disputed territories, arrest of prominent political dissident.	3
Diplomatic talk/ declaration	Diplomatic talks between targeted country and foreign institutions to resolve current geopolitical issues on disputed territories, unilateral declaration of annexation and unlawful claim, declaration of military invasion, declaration of change in military doctrine by targeted country.	8
Military drills/fight	Military drills conducted by targeted country and allies close to disputed territories, foreign countries military displacement in response to targeted country military drills, single military fight, starting of military operation.	19
Military/non-military incident	Military incident with registered casualties and/or effective destruction of property caused by targeted country, unclaimed incident with casualties close to disputed territories.	4
Sanction	Application of sanctions by foreign countries and supernational institutions to individuals, goods and firms from targeted country.	58
Total		93

Hagan, 2014). The last crypto we consider is Tether, a USD-backed stablecoin. The inclusion of Tether, which is also the third most capitalized cryptocurrency, serves the need to understand if our hypothesis holds for a crypto asset with no volatility in prices, that should represent the best alternative for USD, once the fiat currency becomes unavailable after sanctions are in place (Baur & Hoang, 2021; Fantacci & Gobbi, 2020).

We measure traded volume in absolute terms, as the amount of cryptocurrency bought from selling Ruble, and in relative terms, where absolute volume is divided by the overall volume of cryptocurrency traded from relevant fiat currencies such as USD, EUR, JPY, AUD, CNY, AUD and stablecoin like Tether.[6] To guarantee a proper comparison between trades for the different currencies of origin, volumes are expressed in the destination cryptocurrency.

Table 3 provides a description of the variables mentioned above as well as all other variables employed in the empirical analysis.

3. Methodology

3.1. Hypotheses

Our first hypothesis aims at checking whether we observe a major outflow of money from the targeted country domestic currency (i.e., the Ruble) to cryptocurrencies around the release of the news related to the events described in Table 1. Cryptocurrencies in times of geopolitical crisis may serve to provide liquidity, represent store of value and, at the same time, be sufficiently detached by the international system of payments to offer protection from any seizure.

Hypothesis 1. (H1). Cryptocurrencies provide an immediate exit strategy to targeted individuals' assets when sanctions are applied and/or might be applied in the short term.

The empirical analysis also aims at understanding the behavior of targeted individuals in times of war, once the full set of sanctions is in force. In the choice among different cryptocurrencies, targeted individuals may prefer cryptos that provide a higher degree of privacy when they exchange funds under international sanctions. On the other hand, they may dislike cryptos with a lower degree of decentralization of decisions (permissioned protocols) and/or closely tied to fiat currency (fiat-backed stablecoins).

Our second hypothesis relates to the impact of crypto-related sanctions against Russia that the EU implemented with the fight and eight packages of restrictive measures. With the fifth package, enacted on April 8, 2022, the European Commission agrees to a set of restrictive measures regarding, among other things, a prohibition on providing high-value crypto asset services to Russia. Specifically, it is prohibited to provide crypto-asset wallet, account or custody services to Russian nationals or natural persons residing in Russia, or legal persons, entities or bodies established in Russia, if the total value of crypto-assets per wallet, account or custody provider exceeds 10,000 euro (EU, 2022c). With the eight package of restrictive measures the limit of 10,000 euro is removed and the ban applies regardless of the value of the wallet, account or custodian service. However, this ban targets only EU-based crypto-assets custodian services and, probably, EU-based non-custodian wallets providers that performs KYC during the onboarding of the new investor and fails to address any transfer of assets between privately-owned non-custodian wallets (also called un-hosted wallets).[7] Whether the sanctions on crypto-assets are effective or not is relevant for the entire pillar of restrictive measures that refers to freeze of financial assets that the EU applied long before April 2022. The late inclusion of crypto-assets aims at solving any loophole in the implemented

[6] Except for the relative volume for Rubles to Tether that excludes the USD-backed stablecoin from its use in the denominator. So that the relative volume metric for Tether is comparable with the one calculated for the other four cryptocurrencies.

[7] This limit is also pointed out by the EU with respect to the application of the so called "travel rule" to prevent money laundering in the EU via crypto assets transfer between person-to-person un-hosted wallets (EU, 2022d).

Table 2
Timeline of EU sanctions against Russia over Ukraine The Table provides a timeline of the most relevant restrictive measures that the EU adopted against Russia following the illegal annexation of Crimea since 2014. The Table omits sanction time and perimeter extensions for ease of understanding. For a more detailed timeline visit: https://www.consilium.europa.eu/en/policies/sanctions/restrictive-measures-against-russia-over-ukraine/history-restrictive-measures-against-russia-over-ukraine/.

Date	Sanction description
March 3, 2014	The EU and G8 suspend the preparations for the G8 Summit in Sochi, and agreed to work on the adoption of restrictive measures to freeze and recover assets of persons identified as responsible for the misappropriation of Ukrainian state funds.
March 17, 2014	First set of restrictive measures (assets freeze and travel bans) against 21 Russian and Ukrainian officials responsible for actions threatening Ukraine's territorial integrity. Furtherly expanded following the annexation of Crimea and Sevastopol to the Russian Federation.
June 23, 2014	Import ban on goods from Crimea and Sevastopol.
July 16, 2014	Suspension of new financing operations in the Russian Federation by the EIB and by European Bank for Reconstruction and Development. Widening of the legal base for restrictive measures concerning the situation in Ukraine paving the way for imposing asset freezes and visa bans on persons and entities that actively support or are benefiting from Russian decision makers responsible for the annexation of Crimea or the destabilisation of Eastern Ukraine.
July 29, 2014	EU nationals and companies may no more buy or sell new bonds, equity or similar financial instruments with a maturity exceeding 90 days, issued by major state-owned Russian banks, development banks, their subsidiaries outside the EU and those acting on their behalf. Services related to the issuing of such financial instruments, e.g. brokering, are also prohibited. In addition, an embargo on the import and export of arms and related material from/to Russia is imposed that covers all items on the EU common military list.
September 12, 2014	EU nationals and companies may no more provide loans to five major Russian state-owned banks. Trade in new bonds, equity or similar financial instruments with a maturity exceeding 30 days, issued by the same banks, are now prohibited.
December 18, 2014	Investment in Crimea or Sevastopol is outlawed. Europeans and EU-based companies may no more buy real estate or entities in Crimea, finance Crimean companies or supply related services
November 9, 2015	The EU added six members of the Russian Federation State Duma elected from the illegally annexed Autonomous Republic of Crimea and the city of Sevastopol to the list of persons subject to restrictive measures
February 23, 2022	First package of sanctions in response to Russia's invasion of Ukraine. Restrictive measure (assets freeze and travel ban) on all the 351 members of the Russian State Duma. Import ban on goods from the non-government controlled areas of the Donetsk and Luhansk oblasts, restrictions on trade and investments related to certain economic sectors, prohibition to supply tourism services. Export ban for certain goods and technologies and sectoral prohibition to finance the Russian Federation, its government and Central Bank.
February 25, 2022	Second package of sanctions in response to Russia's invasion of Ukraine. Individual sanctions on Vladimir Putin and Sergey Lavrov. Prohibition of listing and provision of services in relation to shares of Russian state-owned entities on EU trading venues, prohibition of accepting deposits exceeding certain values from Russian nationals or residents, holding of accounts of Russian clients by the EU Central Securities Depositories, as well as selling of euro-denominated securities to Russian clients. Export ban on goods and technologies in oil refining. Ban on the sale of all aircrafts, spare parts and equipment to Russian airlines. Diplomats, other Russian officials, and business people no longer benefit from visa facilitation provisions.
February 28, 2022	Third package of sanctions in response to Russia's invasion of Ukraine. Ban on transactions with the Russian Central Bank Support package of 500 million Euro to finance equipment and supplies to Ukrainian armed forces. Ban on the overflight of EU airspance and on access to EU airports by Russian carriers
March 2, 2022	SWIFT ban for seven Russian banks. Ban on investing to future projects co-financed by the Russian Direct Investment Fund. Ban to selling and supplying of euro banknotes to Russia and any related natural or legal person or entity. Suspension of the broadcasting activities in the EU of the outlets Sputnik and Russia Today. On March 9, 2022 similar restriction are applied to Belarus.
March 15, 2022	Fourth package of sanctions in response to Russia's invasion of Ukraine. Ban on all transaction to specific state-owned enterprises. Ban on credit rating services. Ban on new investments in the Russian energy sector and further trade restrictions concerning iron, steel and luxury goods.
April 8, 2022	Fifth package of sanctions in response to Russia's invasion of Ukraine. Ban on imports from Russia of coal and other solid fossil fuels, on Russian vessels from accessing EU ports, on Russian and Belarusian road transport operators from entering the EU. Ban on imports of wood, cement, seafood and liquor. Ban on exports of jet fuel. Ban on deposits to crypto-wallets above 10,000 Euro.
June 3, 2022	Sixth package of sanctions in response to Russia's invasion of Ukraine. Ban on imports from Russia to crude oil and refined petroleum products. SWIFT ban for additional three Russian banks and on Belarusian bank. Further suspension of broadcasting in the EU for three Eussian state-owned outlets.
July 21, 2022	Seventh package of sanctions in response to Russia's invasion of Ukraine. Ban on import of Russian-origin gold and jewellery.
October 6, 2022	Eight package of sanctions in response to Russia's invasion of Ukraine. Introduction of a price cap related to the maritime transport of Russian oil for third countries and additional restrictions on trade and services with Russia. Ban on deposits to crypto-wallets below 10,000 Euro.

restrictive measure. We present our second testable hypothesis below.

Hypothesis 2. (H2). EU sanctions on providing custodian service to Russia hampers the flow from Ruble to cryptocurrencies.

Our third hypothesis refers to the behavior of individuals that are not targeted by the sanctions but anyhow suffer from the military invasion of Ukraine. In particular, we investigate whether Ukrainian citizens increase their reliance on cryptocurrencies after the start of the war. With this third testable hypothesis we intend to study the other side of the battlefront: coin: in Ukraine the war cripples the national banking industry and payment system, and cryptocurrencies offer an alternative, less mature, measure of payment. We present our testable hypothesis below.

Hypothesis 3. (H3). Ukrainian individuals rely more on cryptocurrencies as a measure of payments after the start of the war.

3.2. Econometric approaches

We use three approaches: a panel approach with cryptocurrency fixed effects (to control for heterogeneity across cryptos) and year fixed-effects (to control for the sizeable growth of crypto trading over time); a time series approach (to estimate sensitivity factors that

Table 3
Variables description The Table describes the variables employed in our study for our testable hypothesis in Equation (1). The sample is composed by 7386 daily observations from 2013 to March 2022. We collect volumes on destination cryptocurrencies from Cryptocompare.com, blockchain control variables on Bitcoin and Ethereum networks from Glassnode.com and all other variables are collected from Refintiv.

Variable	Description
RUB	Daily log-returns for USD to Russian Ruble currency exchange rate (USDRUB). Positive (negative) values represent a weakening (strengthening) of Russian Ruble against USD.
UAH	Daily log-returns for USD to Ukrainian Hrvyni currency exchange rate (USDUAH). Positive (negative) values represent a weakening (strengthening) of Ukrainian Hrvyni against USD.
Sp500	S&P 500 daily log-returns.
Crypto	Daily log-returns for the destination cryptocurrency to USD currency exchange rate.
Volume	Overall daily traded volume in the destination cryptocurrency from USD, EUR, JPY, AUD, CNY, AUD, RUB and Tether (USDT). The destination cryptocurrency is the quote currency.
VolRUB	Daily traded volume in destination cryptocurrency from RUB. Destination cryptocurrency is the quote currency.
VolRUBbps	Percentage of daily traded volume in destination cryptocurrency from RUB on the overall volume traded in destination cryptocurrency, calculated as follows $VolRUBbps_t = \frac{VolRUB_t}{Volume_t}$. The variable is represented in basis points (bps) on the overall volume.
ΔVolume	Daily log-difference in Volume.
ΔVolRUB	Daily log-difference in VolRUB.
ΔVolRUBbps	Daily log-difference in VolRUBbps.
Event	Dummy variable with daily observation with a value equal to 1 when an event described in Table 1 occurs and 0 otherwise.
War	Dummy variable to identify the starting of Russian invasion in Urkaine since February 24, 2022 and takes value equal to one until the application of the first crypto-related sanction by the EU on April 8, 2022.
EU crypto sanction 1	Dummy variable to identify the application of the fifth package of sanctions on April 8, 2022, where custodian of crypto assets for value higher than 10,000 euro owned by Russian individuals or entities it is prohibited for EU service providers. The variable takes value of one until the application of the second crypto-related sanction by the EU on October 6, 2022.
EU crypto sanction 2	Dummy variable to identify the application of the eight package of sanctions on October 6, 2022, where custodian of crypto assets owned by Russian individuals or entities it is prohibited for EU service providers regardless of the value in euro.
Account	Log difference in the number of unique addresses that were active in the network either as a sender or receiver. Only addresses that were active in successful transactions are counted. Available for BTC and ETH. The variable is expressed in percentage points.
Fees	Log difference in the total amount of fees paid to miners. Issued (minted) coins are not included. Available for BTC and ETH. The variable is represented in percentage points (pps).
Blocks	Log difference in the number of blocks created and included in the main blockchain in that time period. Available for BTC and ETH. The variable is represented in percentage points (pps).
Supply	Log difference in the total amount of all coins ever created/issued, i.e. the circulating supply. Available for BTC and ETH. The variable is represented in percentage points (pps).
Hash	Log difference in the average estimated number of hashes per second produced by the miners in the network. Available for BTC and ETH. The variable is represented in percentage points (pps).

are crypto-specific); an event study approach. The panel and time series approaches are meant to investigate the long-term impact, whereas the event study approach is intended to look at the short-term impact.

The regression model, which is estimated both as panel and as time series, includes a dummy variable to identify the events described in Table 1. This variable is considered the main explanatory variable to understand traded volume from domestic currency to cryptocurrencies and is used to test for hypothesis 1. Equation (1) presents the econometric model:

$$volrub_{i,t} = \alpha_0 + \alpha_1 event_t + \alpha_2 usdrub_t + \alpha_3 sp500_t + \alpha_4 volume_{i,t} + +\alpha_5 crypto_{i,t} + \Omega control_{i,t} + \varepsilon_t \tag{1}$$

with $volrub_{i,t}$ being a vector of different endogenous variables capturing volume on daily transactions from the domestic currency, the Ruble, to the destination cryptocurrency *i*.

We use four different specifications of volume: log-volume of destination cryptocurrency, Ln(VolRUB), expressed as amount of destination cryptocurrency bought, log-difference of volume of destination cryptocurrency, ΔVolRUB, log-percentage of the overall volume traded in the destination cryptocurrency from Ruble expressed in basis points, Ln(VolRUBbps), and the log-difference of the overall volume traded in the destination cryptocurrency from Ruble expressed in basis points, ΔVolRUBbps.

As for the explanatory variables, $event_t$ is a dummy variable with a value equal to one on the day of the event and zero otherwise,[8] $usdrub_t$ is the log-return on US Dollar to Russian Ruble (USDRUB) exchange rate and represents a proxy of the targeted country currency performance, $sp500_t$ is the log-return on S&P 500 equity index on day t,[9] and $volume_{i,t}$ is the sum of destination cryptocurrency i volume from all the currencies introduced above, comprehensive of volume from Ruble to destination cryptocurrency i, $volume_{i,t}$ is expressed as log-values or log-difference, pairing with the selected endogenous variable from vector $volrub_{i,t}$, $crypto_{i,t}$ is the log-difference in price for cryptocurrency i in time t, $control_{i,t}$ is a vector of five different control variables related to blockchain infrastructure for cryptocurrency i. The control variables represent fundamental values for cryptocurrencies according to Pagnotta and

[8] Instead of using a dummy variable to account for single day events, since the start of the war the $event_t$ variable is substituted with war_t variable, which takes a value of one for any day after the start of the war on February 24, 2022, until the approval of the first crypto-related sanction by the EU on April 8, 2022.

[9] Cryptocurrencies represent both payment methods as well as speculative assets. We included the S&P500 index in the estimated models to control for the overall return from risky assets.

Buraschi (2018), Biais et al. (2020), Pagnotta (2021) and Bhambhwani et al. (2021).

As a complementary approach we perform an event study analysis on our sample of 93 events and test for abnormal spikes in volume from Ruble to cryptocurrency on event days. We perform the analysis both on an event window lasting one day (i.e., the day of the event) and on an event window lasting three days (i.e., from the day prior to the day after the event). We narrowed down the event window to the two most conservative timeframes to exclude confounding effects from close-by events and as a measure of robustness' check for our results.

The cryptocurrency market is still evolving and some patterns observed during early years of the sample might not hold for most recent observations. Additionally, the value of cryptos is quite different over the entire sample period. For this reason, we estimate the baseline regression for the expected volume on a rolling basis. For each j^{th} event we perform a rolling regression using a time series of one year of daily data to identify the expected traded volume from RUB to cryptocurrency on the day of the event, if the j^{th} event window falls under the rolling period for the baseline regression related to the $j^{th}+n$ event, we exclude the entire event window from the timeseries. As most recent observation we use observations in $t-10$ from the day of the event, this to avoid any interference with possible anticipation of the news. Equation (2) present the baseline regression to estimate the expected growth in volume from RUB to cryptocurrency, $\overline{volrub}_{i,t}$.

$$volrub^{j}_{i,t} = \alpha^{j}_{0} + +\alpha^{j}_{1} volume_{i,t} + \varepsilon_t \tag{2}$$

Once estimated the expected growth in volume we can calculate the abnormal growth in volume (AV) as the difference between actual growth in volume $av_{i,t} = volrub_{i,t} - \overline{volrub}_{i,t}$.

To assess the impact of EU sanctions, tested by hypothesis 2, we rely on a regression analysis similar to Equation (1). The model reported as Equation (3) presents the econometric specification we use to test our hypothesis on crypto-sanction effectiveness:

$$volrub_{i,t} = \alpha_0 + \alpha_1 sanction_t + \alpha_2 usdrub_t + \alpha_3 sp500_t + \alpha_4 volume_{i,t} + +\alpha_5 crypto_{i,t} + \Omega control_{i,t} + \varepsilon_t \tag{3}$$

where $sanction_t$ alternatively represents a dummy variable that identifies the introduction of the fifth and eight packages of restrictive measure against Russia.

Lastly, hypothesis 3 tests whether Ukrainian individuals rely more on cryptocurrencies since the start of the war due to the disruption of the Ukrainian system of payments. Equation (4) presents the econometric model we use to test our hypothesis 3:

$$voluah_{i,t} = \alpha_0 + \alpha_1 war_t + \alpha_2 usduah_t + \alpha_3 sp500_t + \alpha_4 volume_{i,t} + +\alpha_5 crypto_{i,t} + \Omega control_{i,t} + \varepsilon_t \tag{4}$$

where $voluah_{i,t}$ being a vector of different endogenous variables capturing log-volume and percentage volume over total transactions on daily transactions from the domestic currency, the Ukrainian Hryvni, to the destination cryptocurrency i, $usduah_t$ represent the log-return on the US dollar Ukrainian Hryvni (USDUAH) exchange rate and $volume_{i,t}$ is the sum of destination cryptocurrency i volume from all the fiat currencies introduced above for Equation (1), comprehensive of volume from Hryvni to destination cryptocurrency i,

4. Results

4.1. Descriptive statistics and correlations

Table 4 shows descriptive statistics for our variables of interest and Table 5 provides pairwise correlation for the entire sample. As previously mentioned, our main variables of interest are the following: the traded volume from RUB to the destination cryptocurrency (VolRUB), the relative weight of traded volume from RUB to the destination cryptocurrency on all traded volume for the destination cryptocurrency (VolRUBbps), and the dummy variable identifying the day the news is released ($event_{i,t}$).

In Table 4 we observe that the cryptocurrency volume increases by 5.3% daily, outflow volume from RUB also increases, however at a more moderate pace of 3.6%, hence Ruble outflows decreases by 1.5% in relative terms. As for the starting relative weight of Ruble outflows, these usually account for approximately 29 basis points, although the deviation from the mean is severe. Section B. of Table 4 provides descriptive statistics on non-event days, here we evidence almost the same results we have in Section A., on the other hand, when we focus only on event days as presented in Section C., we discover an increase in volume from RUB tenfold what we evidence during non-event days, a growth in relative volume by 12.2% with a sixfold increase in the statistical significance.

As for variables correlation, Section A of Table 5 shows positive correlation between event days and growth in outflow of Ruble to cryptocurrency. The evidence is stronger once we include in the analysis the correlation between event days, return on USDRUB, and growth in traded volume for the cryptocurrency. We see that during event days Ruble deteriorates in value and the overall volume in cryptocurrency increases, even with the decrease in purchasing power of the domestic currency, the amount of cryptocurrency bought in exchange for Ruble increases.

Sections B. to F. of Table 5 provide a time series pairwise correlation results for the five cryptocurrencies. As expected, the overall volume traded in Bitcoin decreases with an appreciation in the cryptocurrency, when Bitcoin become more expensive less money flows into the cryptocurrency. If we look at the relative weight of trades from Ruble to Bitcoin (VolRUBbps) the correlation sign with cryptocurrency returns is positive, suggesting that trades from Ruble weights more on the overall market when Bitcoin appreciate and that other fiat currencies see their volume decrease when Bitcoin appreciates. This preliminary evidence suggests a specific behavior for the flow from Ruble to Bitcoin, as the purchasing power of the domestic currency (i.e., the Ruble) does not play a role in the demand for Bitcoin.

Table 4
Descriptive statistics The Table provides descriptive statistics for the variables employed in our study for our testable hypothesis in Equation (1). The sample is composed by 7386 daily observations from 2013 to March 2022. We collect Volumes on destination cryptocurrencies from Cryptocompare.com, we collect control variables on BTC and ETH from Glassnode.com and all other variables from Refintiv. See Table 3 for variables definitions and unit of measure. We include descriptive statistics on volume levels only for VolRUBbps which is, as a relative metrics, the only comparable variable across all five cryptocurrencies. We exclude panel descriptive statistics for Volume and VolRUB because their quote is expressed in cryptocurrency, so the variable spans significantly in value depending on the selected cryptocurrency (from Bitcoin worth thousands of USD, to Tether worth one USD to Ripple worth few cents on USD). The Table provides the first two moments of the distribution, as well the 5th, 25th, 50th, 75th and 95th percentiles.

	N. Obs.	Mean	Std. Dev.	5th pct	25th pct	Median	75th pct	95th pct
A. Full Sample								
RUB	7386	.001	.014	−.012	−.004	0	.004	.014
Sp500	7386	0	.012	−.018	−.003	.001	.006	.015
Crypto	7382	.001	.057	−.072	−.017	0	.019	.08
VolRUBbps	7382	29.068	242.697	.798	2.931	6.087	19.851	101.274
ΔVolume	7382	.053	.398	−.528	−.182	.019	.258	.752
ΔVolRUB	7378	.036	.449	−.597	−.157	.012	.212	.739
ΔVolRUBbps	7230	−.015	.45	−.679	−.214	−.012	.196	.608
Account	3848	2.235	11.025	−13.58	−4.081	1.134	7.531	22.992
Fees	3848	.018	.011	.004	.01	.012	.026	.039
Blocks	3848	5.925	27.897	−27.73	−6.909	3.808	17.435	47.091
Supply	3848	−.15	9.476	−16.814	−3.234	−.095	2.667	15.928
Hash	3848	.213	9.135	−15.374	−3.492	.228	3.81	15.894
B. Non-Event days Sub-sample (Event = 0)								
RUB	7190	.001	.014	−.012	−.004	0	.004	.013
Sp500	7190	0	.012	−.018	−.003	.001	.006	.015
Crypto	7186	.002	.053	−.071	−.017	0	.02	.08
VolRUBbps	7186	29.164	245.892	.773	2.892	5.985	19.536	101.413
ΔVolume	7186	.049	.395	−.532	−.184	.016	.251	.74
ΔVolRUB	7182	.028	.443	−.599	−.16	.007	.205	.711
ΔVolRUBbps	7034	−.019	.447	−.684	−.215	−.013	.192	.589
Account	3727	2.244	10.982	−13.533	−4.076	1.182	7.51	22.985
Fees	3727	.017	.011	.004	.01	.012	.026	.039
Blocks	3727	5.791	27.79	−27.73	−6.875	3.808	17.338	46.238
Supply	3727	−.119	9.447	−16.705	−3.158	−.084	2.65	15.841
Hash	3727	.23	9.11	−15.349	−3.469	.228	3.788	15.865
C. Event days Sub-sample (Event = 1)								
RUB	196	.005	.016	−.01	−.003	.002	.008	.029
Sp500	196	0	.008	−.019	−.003	.001	.006	.012
Crypto	196	−.017	.133	−.172	−.044	−.007	.011	.064
VolRUBbps	196	25.542	40.555	1.425	4.936	9.075	23.402	97.01
ΔVolume	196	.219	.471	−.439	−.121	.153	.593	.996
ΔVolRUB	196	.341	.552	−.394	−.032	.248	.698	1.308
ΔVolRUBbps	196	.122	.515	−.663	−.183	.082	.411	1.003
Account	121	1.958	12.314	−17.871	−4.557	.727	7.883	24.27
Fees	121	.02	.012	.004	.011	.021	.029	.036
Blocks	121	10.053	30.848	−26.93	−7.22	3.066	23.936	57.803
Supply	121	−1.112	10.298	−20.764	−8.054	−.621	4.055	16.363
Hash	121	−.325	9.898	−15.807	−5.587	.234	4.237	16.197

We also performed the Dickey-Fuller unit-root test to check for stationarity in the main variables of interest. All results confirm that the variables are stationary.

4.2. Regression analysis on the outflow from the targeted country currency to cryptocurrencies (Hypothesis 1)

Table 6 presents a set of regressions with year fixed effects and robust standard errors. The first model in Table 6 proves our initial suggestions from Table 5: a negative relationship between Ruble appreciation and outflow of Ruble to cryptocurrencies, regardless of the release of news related to sanctions applications. As the domestic currency loses value against US dollar the cash flow from Ruble to cryptocurrencies increases. With model (2) we introduce $event_{i,t}$ as a new regressor, and we observe that the outflow volume from Ruble to cryptocurrency significantly increases during event days. The finding is true for outflow in absolute and relative terms (see model (3)), and results are consistent if we use log-level instead of log-differences (models (4) and (5)). The last two models of Table 6 provide the analysis since the war started. Here we substitute the variable $event_t$ with war_t as we shift away from an event-based dummy variable to a period-based dummy variable. We also expand the period of the analysis to November 2022. Here we can

Table 5
Pairwise correlation The Table provides pairwise correlations for the panel sample and for individual cryptocurrencies.

Variables	RUB	Sp500	Crypto	ΔVolume	ΔVolRUB	ΔVolRUBbps
A. Full sample						
RUB	1.000					
Sp500	−0.229***	1.000				
Crypto	0.034***	0.104***	1.000			
ΔVolume	0.042***	−0.064***	−0.012	1.000		
ΔVolRUB	0.069***	−0.067***	0.005	0.424***	1.000	
ΔVolRUBbps	0.033***	−0.010	0.011	−0.423***	0.629***	1.000
Event	0.046***	−0.005	−0.053***	0.069***	0.112***	0.051***
B. Bitcoin (BTC)						
RUB	1.000					
Sp500	−0.244***	1.000				
Crypto	−0.006	0.086***	1.000			
ΔVolume	0.037*	−0.051**	−0.101***	1.000		
ΔVolRUB	0.059***	−0.046**	0.015	0.199***	1.000	
ΔVolRUBbps	0.029	−0.003	0.079***	−0.428***	0.782***	1.000
Event	0.023	0.001	0.026	0.026	0.112***	0.082***
C. Ether (ETH)						
RUB	1.000					
Sp500	−0.244***	1.000				
Crypto	0.046*	0.178***	1.000			
ΔVolume	0.043*	−0.067***	−0.043*	1.000		
ΔVolRUB	0.089***	−0.097***	−0.018	0.654***	1.000	
ΔVolRUBbps	0.069***	−0.052**	0.019	−0.224***	0.591***	1.000
Event	0.023	0.001	−0.110***	0.124***	0.149***	0.060**
D. Dash (DASH)						
RUB	1.000					
Sp500	−0.244***	1.000				
Crypto	0.058**	0.087***	1.000			
ΔVolume	0.068**	−0.118***	0.089***	1.000		
ΔVolRUB	0.062**	−0.032	−0.012	0.399***	1.000	
ΔVolRUBbps	−0.004	0.077***	−0.092***	−0.535***	0.561***	1.000
Event	0.023	0.001	−0.168***	0.105***	0.117***	0.013
E. Ripple (XRP)						
RUB	1.000					
Sp500	−0.244***	1.000				
Crypto	0.013	0.173***	1.000			
ΔVolume	0.038	−0.079***	0.087***	1.000		
ΔVolRUB	0.053*	−0.112***	0.014	0.716***	1.000	
ΔVolRUBbps	0.034	−0.072**	−0.076**	−0.086***	0.634***	1.000
Event	0.023	0.001	−0.098***	0.080***	0.101***	0.056*
F. Tether (USDT)						
RUB	1.000					
Sp500	−0.244***	1.000				
Crypto	0.392***	−0.249***	1.000			
ΔVolume	0.037	−0.036	0.045	1.000		
ΔVolRUB	0.093***	−0.052*	0.071**	0.329***	1.000	
ΔVolRUBbps	0.036	−0.005	0.011	−0.698***	0.447***	1.000
Event	0.023	0.001	0.081***	0.063**	0.062**	−0.013

***p < .01, **p < .05, *p < .1.

Table 6
Panel regression analysis This Table provides the estimates of a panel regression analysis for the model presented in Equation (1) with a timespan from January 2013 to November 2022. Models from (1) to (5) estimate Equation (1) and study if variable $event_{i,t}$, which is populated by 93 different single day events related to sanction applications to Russia, provide a valuable explanation to the outflow of Ruble to cryptocurrencies. Models from (6) to (7) verify if the demand for cryptocurrencies endures once the war is started. Here the main explanatory variable is $war_{i,t}$, and takes a value of one for any day after the start of the war on February 24, 2022, and zero otherwise. We include several diagnostic statistics tests: Breusch-Pagan (1979) to check for heteroskedasticity in the sample, Ramsey (1969) RESET test for general specification, D'Agostino et al. (1990) to test for normality in the residuals (the two p-values indicate the test for skewness and kurtosis) and Wooldridge (2002) to test for first-order autocorrelation in the residuals for panel data.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	ΔVolRUB	ΔVolRUB	ΔVolRUBbps	Ln(VolRUB)	Ln(VolRUBbps)	Ln(VolRUB)	Ln(VolRUBbps)
RUB	1.427***	0.821**	0.647**	2.045**	2.045**	2.167**	2.167**
	(5.521)	(3.050)	(3.140)	(3.107)	(3.107)	(3.433)	(3.433)
Crypto	−0.653*	−0.568	−0.568*	−0.059	−0.059	−0.120	−0.120
	(-2.246)	(-2.065)	(-2.576)	(-0.468)	(-0.468)	(-0.908)	(-0.908)
Sp500		−1.124	−0.200	1.019	1.019	0.766	0.766
		(-1.755)	(-0.249)	(2.123)	(2.122)	(1.326)	(1.326)
ΔVolume		0.531**					
		(4.087)					
Event		0.188***	0.113*	0.365***	0.365***		
		(6.187)	(2.184)	(4.938)	(4.938)		
War						0.815*	0.815*
						(2.562)	(2.562)
Ln(Volume)				0.857***	−0.143	0.859***	−0.141
				(6.290)	(-1.052)	(6.354)	(-1.045)
Constant	0.149***	0.091***	0.048***	−8.173***	1.037	−8.188***	1.022
	(25.104)	(5.500)	(9.211)	(-4.655)	(0.591)	(-4.697)	(0.587)
Observations	8077	8077	8077	8077	8077	8077	8077
F-stat	4.37	123.56	2.58	1357.83	409.18	1059.54	1589.77
R-squared	0.016	0.213	0.011	0.645	0.363	0.648	0.370
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust
Year FE	YES	YES	YES	YES	YES	YES	YES
Crypto FE	YES	YES	YES	YES	YES	YES	YES
Breusch-Pagan	0.023	0.000	0.000	0.000	0.000	0.000	0.000
Ramsey RESET	0.922	0.000	0.003	0.000	0.000	0.000	0.000
Normality	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000	0.000/0.000
Wooldridge	0.001	0.004	0.007	0.011	0.011	0.0013	0.0013

t-values are in parentheses
$^{***}p < .01$, $^{**}p < .05$, $^{*}p < .1$

detect whether the same behavior we evidence in previous models stays in war times, as well as if the selling of Ruble for cryptos happens on a more structural and less intense way. Models (6) and (7) show that this behavior from targeted individuals lives on during the war with a higher magnitude. Additionally, as also presented by Fig. 2, most of the trades happen in the early days of war.[10] As a robustness check, we include in Table A. 2 blockchain networks control variables in the analysis for those coins providing the data, Bitcoin and Ether,[11] and evidence on our main testable hypothesis are confirmed: individuals from targeted country exchange Ruble for cryptocurrency in times of liquidity distress due to the application of financial sanctions.[12]

The dummy variable approach assumes that all events are equally important. However, it is possible that the events have different effects. For this reason, we conduct an additional empirical analysis. Specifically, we construct subsamples based on the magnitude of the events and we estimate the baseline model reported in Table 6 using the different subsamples. The subsamples are constructed on

[10] Results from a different model where war_t variable takes value of 1 only for the first week of war evidence a stronger economic and statistical significance for the coefficient, with a coefficient of 1.29 and a t-statistic of 5.80 in both the alternative models (6) and (7). These results are not reported in the current version of the paper and are available from the authors upon request.

[11] It is worth mentioning that blockchain network data for Dash are unavailable in a timeseries format but consultable from dedicated website such as www.blockchair.com. As for Ripple, to the best of our knowledge, blockchain network data are unretrievable in any form, being the coin based on a private permissioned system. Lastly, we have Tether that runs on different blockchain networks; hence the analysis does not apply.

[12] As a robustness check, we also estimated a time series model with lagged values up to one week. Results are reported in Table A. 3 in the Appendix. In the table, for the sake of brevity, we present coefficient results only for the autoregressive component and the main impulse variable 'event'. Nevertheless, the estimated model includes contemporaneous and lagged factors for all the exogenous variables (i.e., the log-return for the cryptocurrency, the log-return for the USDRUB exchange rate and for S&P 500 stock index, the overall log-volume traded from all fiat currency to the cryptocurrency). In terms of results, traded volume from RUB to cryptocurrency (BTC, ETH, DASH, XRP, or USDT) is positively correlated for the entire week of trading. As for the event variable, we find an increase in traded volume up to the next day after the occurrence of the event for BTC, ETH, and XRP.

Table 7

Panel regression analysis with different sub-samples The Table provides panel regression analysis for model presented in Equation (1) with a restricted sample of events based with respect to the results shown in Table 6. Model (1) considers only events that belong to the first quartile in terms of volatility on the RUB exchange rate, whereas Model (2) considers only events that belong to the fourth quartile of the same set. Model (3) considers only sanction events that belong to the first quartile in terms of sanctioned entities/individuals. Model (4) considers only sanction events that belong to the fourth quartile in terms of sanctioned entities/individuals. Model (5) restricts the events sample only to sanctions and Model (6) restricts the events sample only to sanctions that are applied for the first time, no extension of previously declared sanctions is considered. Lastly, Model (7) restricts the events sample only to the 19 events that regard military drills or fights.

Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)
RUB	0.904**	0.782*	0.904**	0.822**	0.900**	0.898**	0.809**
	(3.355)	(2.661)	(3.359)	(3.009)	(3.322)	(3.301)	(3.042)
Crypto	−0.593*	−0.583	−0.596*	−0.563	−0.588*	−0.594*	−0.585*
	(-2.253)	(-2.131)	(-2.275)	(-2.030)	(-2.215)	(-2.234)	(-2.199)
Sp500	−1.086	−1.129	−1.085	−1.115	−1.097	−1.078	−1.104
	(-1.687)	(-1.748)	(-1.687)	(-1.738)	(-1.711)	(-1.672)	(-1.716)
ΔVolume	0.535**	0.530**	0.535**	0.530**	0.535**	0.535**	0.531**
	(4.094)	(4.083)	(4.096)	(4.084)	(4.102)	(4.110)	(4.088)
Event	0.045	0.290***	−0.064	0.209***	0.119**	0.228***	0.327**
	(1.707)	(9.403)	(-1.947)	(9.395)	(4.244)	(6.095)	(4.301)
Constant	0.097***	0.099***	0.098***	0.090***	0.098***	0.098***	0.092***
	(5.884)	(5.872)	(5.875)	(5.365)	(5.848)	(5.854)	(5.501)
Observations	8077	8077	8077	8077	8077	8077	8077
R-squared	0.210	0.212	0.210	0.213	0.211	0.211	0.213
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust
Year FE	YES	YES	YES	YES	YES	YES	YES
Crypto FE	YES	YES	YES	YES	YES	YES	YES
Events sample	Low volatility RUB	High volatility RUB	Less severe sanctions	More severe sanctions	Sanctions	No Extension	Fights/Drills

t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

the basis of three criteria and the estimates are reported in Table 7.

The first criterion focuses on the market impact of the events. We organize the full sample (made of 93 events) in terms of absolute log-difference in the exchange rate between RUB and USD and we attribute to each event the relative weight in terms of overall volatility produced. We estimate the weight as follows:

$$weight_t = \frac{|usdrub_t|}{\sum_{i=1}^{N} |usdrub_t|}$$

where $|usdrub_t|$ is the absolute return on USDRUB exchange rate for the t event and N represent the overall number of 93 events. Then, we select only ‘low volatility’ events included in the first quartile of variable $weight_t$ (see model (1)), and only ‘high volatility’ events included in the fourth quartile of variable $weight_t$ (see model (2)). We find that ‘low volatility’ events carry no significance in terms of outflow of RUB to cryptocurrencies, whereas ‘high volatility’ events increase volume from RUB to cryptocurrencies by 29%.

The second criterion focuses on the number of entities and individuals sanctioned in each of the 58 events related to sanctions. We collect from https://www.opensanctions.org/datasets/sanctions/the variable $size_t$ that represents the number of entities and individuals sanctioned in each of the 58 sanction events. Then, we select only ‘less severe’ sanctions included in the first quartile of variable $size_t$ (see model (3)), and only ‘more severe’ sanctions included in the fourth quartile of variable $size_t$ (see model (4)). We find that ‘less severe’ sanctions carry no significance in terms of outflow of RUB to cryptocurrencies, whereas ‘more severe’ sanctions increase volume traded from RUB to cryptocurrencies by 20%.

The third criterion simply selects a subset of events based on their category. In model (5) we consider only events that relate to the application of sanctions whereas in model (6) we further restrict the subset of events and select only newly declared sanctions and exclude events where the authorities extend previously approved sanctions. When only newly declared sanctions are considered the increase in volume from RUB to cryptocurrencies is almost twofold (23% vs 12%). Finally, in model (7) we only consider the 19 events that relate to military drills or fights. This last model shows the highest relevance in terms of outflow of fiat currency to cryptocurrency as the volume from RUB increases by 33%.

Next, we use a time series framework to verify if individuals have preferences for specific cryptocurrency features or they only value their liquidity. We expand our analysis for two reasons: first, we look at the five cryptocurrencies individually to provide robustness check on our results from the panel framework. Second, looking at different cryptos helps understanding whether targeted individuals consider cryptocurrency idiosyncratic features when making their trades. Table 8 shows the results on the analysis related to single day events up to the beginning of the war and Table 9 shows the results when we consider the sample period from the start of the war. For ease of reading, we provide results only for the two most relevant dependent variables: growth in outflow of Ruble, ΔVolRUB (models from (1) to (5)) and its relative weight, ΔVolRUBbps (models from (6) to (10)). From a robustness check perspective, time series analysis confirms our findings that almost all five cryptocurrencies provide a “safe haven” for targeted individuals on event days, as volume increases in absolute and relative terms. About our second proposition, we find that not all five cryptocurrencies are equally interested by the growth in volume.

Table 8
Time series regression analysis The Table provides time series regression analysis for model presented in Equation (1) and for each cryptocurrency.

Dependent variable	ΔVolRUB					ΔVolRUBbps				
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cryptocurrency	BTC	ETH	DASH	XRP	USDT	BTC	ETH	DASH	XRP	USDT
RUB	1.404**	0.818	0.590	0.132	0.722	1.404**	0.682	0.483	0.130	0.567
	(2.212)	(1.314)	(1.191)	(0.132)	(0.724)	(2.212)	(1.071)	(0.741)	(0.129)	(0.583)
Crypto	−0.989***	0.190	−0.128	−0.352**	−0.856	−0.989***	0.224	−0.425**	−0.367**	−0.942
	(-4.162)	(0.812)	(-0.837)	(-2.175)	(-0.778)	(-4.162)	(0.979)	(-2.381)	(-2.305)	(-0.714)
Sp500	−0.710	−2.124*	0.628	−2.196*	−2.044*	−0.710	−1.737	2.644***	−2.106	−1.213
	(-0.789)	(-1.794)	(1.137)	(-1.670)	(-1.756)	(-0.789)	(-1.485)	(3.463)	(-1.613)	(-0.957)
ΔVolume	0.342***	0.790***	0.358***	0.967***	0.313***	−0.658***				
	(8.659)	(28.107)	(12.641)	(30.404)	(11.559)	(-16.651)				
Event	0.239***	0.218***	0.159*	0.100*	0.088	0.239***	0.155**	0.002	0.093	−0.060
	(4.150)	(2.954)	(1.867)	(1.658)	(1.219)	(4.150)	(2.165)	(0.021)	(1.573)	(-0.668)
Constant	0.130	−0.034	−0.006	−0.003	−0.035	0.130	−0.040	0.007	−0.007	−0.069
	(0.484)	(-1.037)	(-0.181)	(-0.065)	(-0.745)	(0.484)	(-1.141)	(0.174)	(-0.162)	(-0.885)
Observations	2348	1704	1390	1308	1327	2348	1704	1390	1308	1327
R-squared	0.107	0.446	0.150	0.469	0.141	0.201	0.011	0.014	0.017	0.006
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Crypto FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Breusch-Godfrey LM	273.333	109.570	92.775	110.195	131.816	273.347	109.570	92.775	110.195	131.816
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

t-values are in parentheses.
***$p < .01$, **$p < .05$, *$p < .1$.

Table 9

Time series regression analysis on Russian invasion The Table provides time series regression analysis for model presented in Equation (1) and for each cryptocurrency with a dummy variable 'War' that identifies only observations since the start of the Russian invasion in Ukraine on February 24, 2022 until the application of first crypto-related sanctions on April 8, 2022.

Dependent variable	Ln(VolRUB)					Ln(VolRUBbps)				
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cryptocurrency	BTC	ETH	DASH	XRP	USDT	BTC	ETH	DASH	XRP	USDT
RUB	3.140***	1.221	1.534**	0.468	3.407***	3.140***	1.221	1.534**	0.468	3.407***
	(2.762)	(1.199)	(2.341)	(0.441)	(2.740)	(2.762)	(1.199)	(2.341)	(0.441)	(2.740)
Crypto	0.171	0.083	0.072	0.108	0.783	0.171	0.083	0.072	0.108	0.783
	(0.681)	(0.225)	(0.336)	(0.373)	(0.550)	(0.681)	(0.225)	(0.336)	(0.373)	(0.550)
Sp500	0.317	0.432	0.836	−1.383	−1.430	0.317	0.432	0.836	−1.383	−1.430
	(0.233)	(0.176)	(1.058)	(-0.735)	(-1.130)	(0.233)	(0.176)	(1.058)	(-0.735)	(-1.130)
Ln(Volume)	0.152***	0.525***	0.296***	0.721***	0.463***	−0.848***	0.525***	0.296***	0.721***	0.463***
	(5.148)	(16.259)	(13.682)	(26.166)	(15.574)	(-28.808)	(16.259)	(13.682)	(26.166)	(15.574)
War	0.646***	1.084***	−0.091	1.291***	0.862***	0.646***	1.084***	−0.091	1.291***	0.862***
	(4.149)	(9.650)	(-1.220)	(7.523)	(6.364)	(4.149)	(9.650)	(-1.220)	(7.523)	(6.364)
Constant	−1.367***	−0.468	2.239***	−1.773***	3.588***	7.843***	−0.468	2.239***	−1.773***	3.588***
	(-3.831)	(-1.037)	(8.726)	(-3.299)	(8.366)	(21.974)	(-1.037)	(8.726)	(-3.299)	(8.366)
Observations	2348	1704	1390	1308	1327	2348	1704	1390	1308	1327
R-squared	0.821	0.448	0.376	0.580	0.953	0.771	0.448	0.376	0.580	0.953
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust
Year dummy	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

Table 8 shows that Bitcoin and Ether, the two most popular and liquid coins, attract most of the domestic currency outflow. With Dash, we find positive and significant growth in absolute volume but no growth in relative terms. On top of that, absolute volume growth for Dash is less than what observed for Bitcoin and Ether, the opposite of what we would have expected if users prefer privacy-oriented cryptocurrencies to others. Dash protocol allows users to decide whether to hide or not the origin of their funds, thus making the trade anonymous. Bitcoin and Ethereum networks do not provide any kind of privacy and the opposite is true. Blockchain works in a way that everyone can see the entire trading history for a specific address or coin, and the false idea of anonymity stems from the use of an apparently random-generated string of text to identify the user instead of his/her real name. However, once we match the string to a specific individual, blockchain networks supposed anonymity backfires, and the network proves to be way more transparent than old-style fiat currency transactions, which are not registered on a publicly available distributed ledger. This initial comparison between mainstream cryptocurrencies and Dash is not consistent with the idea of a stronger reliance on privacy-oriented coins for targeted individuals. A similar finding emerges when we consider the demand of Dash during war time in Table 9.

We also consider cryptocurrencies where the decentralization feature is not as strong as for Bitcoin and Ether. Ripple has a permissioned network where the blockchain network is managed by a private company that runs a collection of private computer nodes to validate transactions. On Ripple protocol, every transaction is processed by a validator who has special permission to help maintain the Ripple payment network, whereas with Bitcoin, Ether and other permissionless protocols it is possible for anyone to help validate transactions, and impossible for a single user to manipulate the ledger and reject a transaction. An individual eager to step away from currencies where a third party has the power to shut down their trade, would avoid the use of coins such as Ripple. Findings in Table 8 shows that this is not the case, as traded volume from Ruble to Ripple significantly increases during event days, although always less than Bitcoin and Ether. Stronger evidence on Ripple emerges from the analysis presented in Table 9, since the start of the war Ripple is the single most bought cryptocurrency of the sample, probably underlining a certain a preference of targeted individuals to rely on a fast, cheap and yet permissioned, system of payments when cryptocurrencies are considered.

Lastly, we consider a different form of hampered decentralization with stablecoins. With Tether, the company behind the project backs every USDT with one US dollar (or little less). In this case, the cryptocurrency is permissionless and the individual is free to exchange from Ruble to Tether. If we stop at this feature, Tether is the perfect substitute for USD when targeted individuals are prohibited to buy fiat currency. However, Tether is valued on the amount of USD that the company holds in reserves, hence any limitation in these holdings would result in a sudden drop of value for the cryptocurrency. With Tether, our hypothesis holds only when single day events are considered, as we evidence that volume from Ruble to Tether does not significantly increase during event days. However, when we study the use of this stablecoin during the war in Table 9 we notice a strong increase in the exchange of Ruble for Tether, probably targeted individuals are not considering Tether as a kind of sanction sensitive cryptocurrency but as a proper substitute for the USD.

Table 10

Event study analysis on full sample The Table provides descriptive statistics from the event study analysis on 93 events where news relating to the application of sanctions for the targeted country. Event study analysis is conducted on growth in absolute volume exchanged from RUB to the destination cryptocurrency (ΔVolRUB). For each event we estimate expected ΔVolRUB using a one year rolling regression based on Equation (2). Abnormal growth in volume (AV) is calculated as the difference between the observed value for variable ΔVolRUB and the estimated one. Number in brackets represent descriptive statistics for t-statistics.

A. Bitcoin (BTC)								
Event window	N	Mean	SD	Min	p25	Median	p75	Max
Event days (AV)	93	0.137	0.693	−4.148	−0.123	0.165	0.502	1.875
		(6.452)	(21.530)	(-62.994)	(-4.736)	(7.545)	(14.150)	(121.879)
−1/+1 (CAV)	93	0.056	0.511	−1.885	−0.181	0.031	0.318	1.677
		(3.775)	(13.915)	(-15.885)	(-3.038)	(0.518)	(6.162)	(94.849)
B. Ether (ETH)								
Event days (AV)	54	0.126	0.439	−0.776	−0.061	0.118	0.335	1.576
		(5.028)	(15.770)	(-24.473)	(-2.418)	(4.794)	(11.586)	(54.794)
−1/+1 (CAV)	54	0.026	0.541	−1.259	−0.375	0.089	0.249	1.435
		(1.141)	(11.886)	(-25.544)	(-7.917)	(1.777)	(7.588)	(28.796)
C. Dash (DASH)								
Event days (AV)	38	0.173	0.456	−0.460	−0.048	0.026	0.306	1.833
		(7.858)	(19.215)	(-21.464)	(-2.386)	(1.043)	(16.631)	(69.834)
−1/+1 (CAV)	38	0.131	0.527	−0.632	−0.103	0.065	0.234	2.308
		(3.333)	(12.890)	(-14.487)	(-3.534)	(1.699)	(6.164)	(50.976)
D. Ripple (XRP)								
Event days (AV)	36	0.097	0.288	−0.430	−0.065	0.026	0.352	0.679
		(4.558)	(17.011)	(-23.231)	(-4.394)	(1.433)	(16.278)	(46.680)
−1/+1 (CAV)	36	0.047	0.338	−0.671	−0.159	−0.042	0.217	0.927
		(2.114)	(12.054)	(-21.000)	(-4.053)	(-1.326)	(7.528)	(36.678)
E. Tether (USDT)								
Event days (AV)	37	0.081	0.342	−0.863	−0.170	0.106	0.387	0.582
		(4.953)	(15.071)	(-27.295)	(-5.381)	(5.457)	(18.782)	(29.875)
−1/+1 (CAV)	37	0.224	0.467	−0.639	−0.080	0.190	0.586	1.282
		(6.484)	(12.775)	(-13.505)	(-1.125)	(5.443)	(18.459)	(36.433)

4.3. Event study on the outflow from the targeted country currency to cryptocurrencies (Hypothesis 1)

Table 10 shows results for our event study analysis. The Table presents aggregated results for all the available events on all five cryptocurrencies. The only cryptocurrency with data available for all 93 events is Bitcoin, as most of the other coins did not exists at the beginning of the selected time frame. For each cryptocurrency we perform the analysis with two different event windows, in one we apply the analysis on $av_{i,t}$ on the day of the event (an event window of one day) and in the second one we consider an event window of three days (−1/+1 days on event day) and presents results for cumulated abnormal growth in volume, $cav_{i,t}$. As highlighted from our regression analysis Bitcoin is the cryptocurrency that bought mostly during event days, whereas Ether and Ripple show less prominent spike in trading. In opposition to what we discover in our regression analysis Dash and Tether are considered by targeted investors when they need to exchange Ruble.

In Table 11 we perform the same analysis as in Table 10 only for positive $av_{i,t}$ and $cav_{i,t}$. By analysing a sub-sample of positive abnormal volumes, we narrow down the study and we account for possible overshooting of the estimation model we present in Equation (2). By doing that we stress the model to evidence if observations with an actual abnormal increase in volume from Ruble to cryptocurrency carry a statistical significance. Results confirm our findings presented in Table 10.

4.4. Effectiveness of EU sanctions on cryptocurrencies (Hypothesis 2)

Hypothesis 2 studies the effectiveness of the solutions implemented by the EU to halt cryptocurrencies trading for Russian individuals, and to close any loophole in the asset-freeze component of the introduced sanctions. As a reminder, the EU agreed on April 8, 2022 to prohibit any EU-based cryptocurrency custodian service provider to allow Russian individuals to detain more than 10,000 euro in crypto asset on their platform. The threshold of 10,000 euro is then removed on October 6, 2022, where custodian of crypto asset is prohibited regardless of the representative value.

Table 12 shows the results for our panel regression analysis. Since April 8, 2022, the trading in cryptocurrency reduces symmetrically, size and statistical significance, to the increase that happened after the start of the war (see Table 6). However, the second sanction application on October 6, does not seem to drive the trading further down. With Table 13 we provide an analysis on a time series framework and investigate if some cryptocurrencies are more influenced than other. Although many factors may play a role, given the complex situation, we find a stronger significance for the second crypto-related sanction for all coins expect for Dash. For ease of read in Table 13 we provide only the coefficients associated with the dummy variable of study, the specification of the model is the same of the one we provide in Table 9. To provide a complete comprehension of the phenomena we also report war_t coefficient value from Table 9. Bitcoin, Ether and Tether, the three most liquid cryptocurrencies, behave similarly. All three present a strong increase in trading after the beginning of the war, and a symmetrical reduction in the trading after the application of the sanctions. On the

Table 11
Event study analysis on positive AV/CAV The Table provides descriptive statistics from the event study analysis on 93 events where news relating to the application of sanctions for the targeted country. Results are presented only for positive values for AV and CAV. Event study analysis is conducted on growth in absolute volume exchanged from RUB to the destination cryptocurrency (ΔVolRUB). For each event we estimate expected ΔVolRUB using a one year rolling regression based on Equation (2). Abnormal growth in volume (AV) is calculated as the difference between the observed value for variable ΔVolRUB and the estimated one. Number in brackets represent descriptive statistics for t-statistics.

A. Bitcoin (BTC)								
Event window	N	Mean	SD	Min	p25	Median	p75	Max
Event days (AV)	63	0.482	0.376	0.003	0.197	0.422	0.621	1.874
		(17.412)	(18.845)	(0.152)	(8.566)	(11.996)	(23.878)	(122.137)
−1/+1 (CAV)	71	0.343	0.398	−0.963	0.085	0.247	0.530	1.737
		(12.586)	(19.652)	(0.116)	(2.476)	(6.550)	(15.166)	(106.771)
B. Ether (ETH)								
Event days (AV)	36	0.332	0.333	0.010	0.116	0.216	0.435	1.576
		(12.587)	(11.878)	(0.533)	(4.264)	(8.603)	(17.027)	(54.794)
−1/+1 (CAV)	35	0.381	0.364	0.028	0.152	0.236	0.426	1.435
		(9.025)	(7.997)	(0.499)	(3.971)	(6.380)	(11.738)	(28.796)
C. Dash (DASH)								
Event days (AV)	23	0.435	0.463	0.024	0.248	0.298	0.521	1.833
		(19.786)	(18.029)	(0.886)	(8.381)	(15.964)	(27.132)	(69.834)
−1/+1 (CAV)	27	0.363	0.548	0.013	0.075	0.168	0.459	2.308
		(9.662)	(12.507)	(0.304)	(2.514)	(4.471)	(11.340)	(50.976)
D. Ripple (XRP)								
Event days (AV)	22	0.305	0.196	0.004	0.176	0.304	0.438	0.679
		(16.423)	(12.746)	(0.324)	(6.181)	(16.098)	(23.323)	(46.680)
−1/+1 (CAV)	21	0.358	0.240	0.144	0.182	0.303	0.555	0.927
		(12.931)	(9.550)	(5.252)	(6.538)	(7.859)	(17.301)	(36.678)
E. Tether (USDT)								
Event days (AV)	23	0.319	0.173	0.015	0.160	0.376	0.465	0.582
		(15.765)	(9.063)	(0.418)	(6.368)	(16.269)	(21.589)	(29.875)
−1/+1 (CAV)	29	0.442	0.361	0.018	0.177	0.258	0.638	1.282
		(12.510)	(9.761)	(0.601)	(5.191)	(8.821)	(20.355)	(36.433)

Table 12
Panel regression analysis on EU crypto-related sanctions applications The Table provides panel regression analysis for model presented in Equation (3).

Variable	(1)	(2)	(3)	(4)
	Ln(VolRUB)	Ln(VolRUBbps)	Ln(VolRUB)	Ln(VolRUBbps)
RUB	1.991**	1.991**	2.244**	2.244**
	(3.028)	(3.028)	(3.563)	(3.563)
Crypto	−0.120	−0.120	−0.108	−0.108
	(-0.922)	(-0.922)	(-0.806)	(-0.806)
Sp500	1.017	1.017	1.210*	1.210*
	(2.075)	(2.075)	(2.752)	(2.752)
Ln(Volume)	0.863***	−0.137	0.857***	−0.143
	(6.486)	(-1.030)	(6.292)	(-1.049)
EU crypto-sanction 1	−0.878*	−0.878*		
	(-2.509)	(-2.509)		
EU crypto-sanction 2			−0.580	−0.580
			(-2.092)	(-2.092)
Constant	−8.241***	0.969	−8.167***	1.043
	(-4.803)	(0.565)	(-4.649)	(0.594)
Observations	8077	8077	8077	8077
R-squared	0.653	0.379	0.645	0.363
Std. Errors	Robust	Robust	Robust	Robust
Year dummy	YES	YES	YES	YES

t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

Table 13
Effects of EU crypto-related sanctions – time series analysis The Table provides a series of individual coefficient and t-statistic for time series regression analysis for model presented in Equation (1) when different dummy variables are considered. Models from 1.a to 5.a present the coefficient and the t-statistic for the dummy variable 'War' that identifies only observations since the start of the Russian invasion in Ukraine on February 24, 2022 to the first application of crypto-assets related sanctions. Models from 1.b to 5.b. present the coefficient and the t-statistic for the dummy variable 'EU crypto sanction 1'that identifies only observations since the first application of crypto-assets related sanctions on April 8, 2022 to the second application of crypto-assets related sanctions. Models from 1.c to 5.c. present the coefficient and the t-statistic for the dummy variable 'EU crypto sanction 2'that identifies only observations since the second application of crypto-assets related sanctions on October 6, 2022. Each pair of coefficient and t-statistic correspond to a specification with RUB, Crypto, Sp500, and Ln(Volume) as control variables and time fixed effects.

Model	(1.a, .b, .c)	(2.a, .b, .c)	(3.a, .b, .c)	(4.a, .b, .c)	(5.a, .b, .c)
Cryptocurrency	BTC	ETH	DASH	XRP	USDT
War	0.646***	1.084***	−0.091	1.291***	0.862***
	(4.149)	(9.650)	(-1.220)	(7.523)	(6.364)
EU crypto sanction 1	−0.597***	−1.337***	0.126***	−1.365***	−0.689***
	(-6.575)	(-19.001)	(3.006)	(-14.283)	(-7.605)
EU crypto sanction 2	−0.624***	−1.045***	0.059	−0.630***	−0.549***
	(-6.998)	(-12.286)	(0.743)	(-3.983)	(-3.052)
Observations	2348	1704	1390	1308	1327
Control Variables	YES	YES	YES	YES	YES
Std. Errors	Robust	Robust	Robust	Robust	Robust
Year dummy	YES	YES	YES	YES	YES

t-values are in parentheses.
****p < .01, **p < .05, *p < .1.*

contrary, Dash does not seem to follow any behavior from war and sanction-related events. On the other hand, the behavior of Ripple emulates the one of Bitcoin, Ether and Tether but with greater magnitude. Since the war started, targeted individuals rely more on cryptocurrency that advocate itself as a revolutionary system of payments, and the same cryptocurrency suffers the most in terms of traded volume once the crypto-related sanctions are introduced.

4.5. *Use of cryptocurrencies in economies with crippled system of payments (Hypothesis 3)*

Hypothesis 3 focuses on the reliance on cryptocurrencies from Ukrainian individuals during times of war. Our goal is understanding if cryptocurrencies provide a valuable alternative to fiat currency when the traditional system of payment is disrupted due to the Russian invasion. To test this hypothesis we rely on the model presented in Equation (4) and Table 14 and Table 15 present our findings where outflow volume from Ukrainian Hrvyni to destination cryptocurrencies is the variable of study and war_t is the explanatory

Table 14
–Panel regression analysis on Ukrainian Hryvni The Table provides panel regression analysis for model presented in Equation (1) with Ukrainian Hrvyni (UAH) instead of Russian Ruble where the aim is to verify if the demand for cryptocurrencies endures once the war is started also for Ukrainian individuals. In this Table the main explanatory variable is $war_{i,t}$, and takes a value of one for any day after the start of the war on February 24, 2022, and zero otherwise.

Variable	(1)	(2)
	Ln(VolUAH)	Ln(VolUAHbps)
UAH	−0.954**	−0.954**
	(-4.171)	(-4.171)
Crypto	−1.384*	−1.384*
	(-2.286)	(-2.286)
Sp500	0.063	0.063
	(0.075)	(0.075)
Ln(Volume)	0.605***	−0.395**
	(5.895)	(-3.849)
War	0.721**	0.721**
	(2.915)	(2.915)
Constant	−5.453**	3.758**
	(-4.345)	(2.994)
Observations	5465	5465
R-squared	0.425	0.370
Std. Errors	Robust	Robust
Year FE	YES	YES

t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

variable. Table 14 presents our results on a panel framework and we evidence a strong increase in outflow since the war started, as Ukrainians migrate to an alternative, yet immature, system of payments and exchange their domestic currency for crypto assets.[13] In Table 15 we expand our analysis on a time series, and it is useful to read our results together with Table 9 and our results on Russian Ruble. Since the start of the war Bitcoin and Tether are the two single most bought cryptocurrencies, where the first one can be perceived as a store of value and the second one as a proxy to trade with US Dollars when the traditional system of payment is crippled. Ether and Dash also provide a strong increase in trading, whereas Ripple show opposite results with respect to the other four and with respect to Ripple findings for Ruble. With Ukrainian national system of payment in peril a cryptocurrency that is meant as a banking-friendly coins probably lacks the institutions infrastructure to play its role. On the contrary, in Russia where we the national system of payments works, buying Ripple when US Dollar is unavailable is the second-best solution for targeted individuals.

5. Conclusions and policy implications

In this paper we examine the use of cryptocurrencies in times of geopolitical crises. Cryptocurrencies promote the decentralization of finance as their more prominent feature and, given the popularity achieved in recent years, they sufficiently represent a liquid currency to deposit money in the short-term and to find an alternative way when a fiat currency is not a viable option. Crypto currencies may be used when the domestic financial system is no longer available as well as to try to escape the seizure of assets during the application of financial sanctions. We also study whether cryptocurrency features other than liquidity are considered by traders, and if certain coins represent a better alternative than others.

The empirical analysis is based on a sample of 93 events associated to potential limitations of a fiat currency circulation and that considers five cryptocurrencies (Bitcoin, Ether, Ripple, Dash, and Tether). We find that trading in crypto currencies increases with events of geopolitical tensions. Bitcoin and Ether - the two most liquid cryptocurrencies in our sample – attract most of the trades. Looking at the most recent period, in association with the Russian invasion of Ukraine, the devaluation of the Russian Ruble generally increases the outflow to crypto currencies, which is consistent with the idea that cryptos were used as an exit strategy from a currency that was losing its value. For the stablecoin Tether, no effect is associated to the 'tensions' event days and – interestingly – a statistically significant relation with the Ruble devaluation. Anecdotal evidence suggests that traders went out from Ruble due to its devaluation, and preferred stablecoins to avoid taking on the additional risk of (what were supposed to be) more volatile crypto currencies. We also study if the reliance on cryptocurrencies increases since the Russian invasion started in February 2022, and we find strong results for all five cryptocurrencies but especially for coins that mostly act as a mean of payment.

The increase in crypto currencies trading in times of crises can be motivated by different explanations (e.g., protecting savings as the domestic currency devaluates, making payments as the domestic financial system is no longer available, avoiding sanctions). We

[13] It is worth mentioning that our analysis does not relate, nor it is affected by the donation of cryptocurrencies that Ukraine received from abroad. Those donated funds are outflow of money from other fiat currencies different than Ukrainian Hryvni to destination cryptocurrencies, then transferred to Ukrainian wallets. Conversely, we rely on outflow of money from Ukrainian domestic currency to cryptocurrencies.

Table 15

Time series regression analysis on Ukrainian Hryvni The Table provides time series regression analysis for model presented in Equation (1) and for each cryptocurrency with a dummy variable 'War' that identifies only observations since the start of the Russian invasion in Ukraine on February 24, 2022.

Dependent variable	Ln(VolUAH)					Ln(VolUAHbps)				
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Cryptocurrency	BTC	ETH	DASH	XRP	USDT	BTC	ETH	DASH	XRP	USDT
UAH	−0.821	1.547	−0.311	2.465	2.925	−0.821	1.547	−0.311	2.465	2.925
	(-0.334)	(0.504)	(-0.030)	(0.162)	(0.598)	(-0.334)	(0.504)	(-0.030)	(0.162)	(0.598)
Crypto	−0.794**	0.542**	−0.690	−2.494	−1.718	−0.794**	0.542**	−0.690	−2.494	−1.718
	(-2.501)	(2.167)	(-1.079)	(-1.290)	(-1.036)	(-2.501)	(2.167)	(-1.079)	(-1.290)	(-1.036)
Sp500	−1.363	−0.421	−4.498	2.207	−0.942	−1.363	−0.421	−4.498	2.207	−0.942
	(-1.185)	(-0.355)	(-1.215)	(0.504)	(-0.720)	(-1.185)	(-0.355)	(-1.215)	(0.504)	(-0.720)
Ln(Volume)	0.201***	0.447***	−0.399***	1.430***	0.541***	−0.799***	−0.553***	−1.399***	0.430***	−0.459***
	(5.026)	(16.479)	(-4.944)	(10.951)	(25.421)	(-19.989)	(-20.422)	(-17.327)	(3.294)	(-21.552)
War	1.598***	0.957***	0.889***	−0.648***	1.197***	1.598***	0.957***	0.889***	−0.648***	1.197***
	(18.102)	(11.027)	(7.958)	(-3.799)	(9.678)	(18.102)	(11.027)	(7.958)	(-3.799)	(9.678)
Constant	−2.829***	−1.156***	6.552***	−20.543***	3.453***	6.381***	8.054***	15.762***	−11.332***	12.664***
	(-7.408)	(-2.901)	(6.127)	(-7.416)	(12.746)	(16.707)	(20.214)	(14.740)	(-4.091)	(46.741)
Observations	1729	1136	907	955	738	1729	1136	907	955	738
R-squared	0.771	0.527	0.510	0.393	0.917	0.844	0.530	0.542	0.218	0.769
Std. Errors	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust	Robust
Year dummy	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

perform a specific analysis with respect to the EU sanctions established in 2022 on Russia (EC (2022), EC (2022a), EC (2022b)). For the full sample, we find that crypto trading slows down when crypto-related services (wallet, account or custody services) are explicitly included in EU financial sanctions packages. When we separately look at the different crypto currencies in our sample, we find split results. For Bitcoin, Ether, Tether and Ripple, a strong increase in crypto trading activity after the beginning of the war is followed by a symmetrical reduction in trading after the application of the sanctions. The symmetrical reduction observed with the ban is consistent with the use of crypto to circumvent sanctions prior to the explicit prohibition on crypto-asset services (wallet, account or custody services) above 10,000 euros which was part of the 4th EU sanctions package enacted in April 2022. For Dash (a more privacy-oriented coin) only, an increase in crypto trading activity in association with the April 2022 EU measures, and no impact of the October 2022 EU extension to services below 10,000 euros.

Our findings suggest that cryptocurrencies, despite being classified as a speculative asset under the Commodity Exchange Act (CFTC, 2022), function as a store of value under extreme conditions (e.g., the imposition of financial sanctions and the limitation of free circulation of domestic currency abroad) as well as a measure of payment since the start of the war. Additionally, prior to the bans on providing custodian services to Russian depositors introduced by the EU, targeted individuals and entities were using liquid cryptocurrencies – at least partly – as a way to avoid financial sanctions and keep their investments into a currency which is internationally tradable. The effectiveness of the bans helps in building a good reputation for cryptocurrencies in the international community.

We also examine trading activity from the Ukranian Hrvynia to crypto currencies and find a strong increase in outflow from Hrvynia since the beginning of the conflict. This finding – which is not affected by by donations of crypto currencies received by Ukraine from abroad – is consistent with the hypothesis that Ukrainians increasingly exchanged their domestic currency for crypto currencies as their domestic fiat currency suffers from severe war time limitations (Tassev, 2022) and a crippled system of payments.

Our combined evidence from Russia and Ukraine shows that crypto currencies deploy significant value in war times. This feature is usually associated to safe haven assets such as gold, except that cryptocurrencies lack of the usual low volatility of a true safe haven. Blockchain networks offer the possibility of fast and decentralized worldwide transactions. This makes cryptocurrencies a credible mean of payment for war time economics. Investors know that the liquidity and transactions cost for a Bitcoin is not comparable to those for developed countries fiat currency but the independence from international payment system and foreign central banks outweighs them in times of geopolitical crises. In this particular state of the world, decentralization is worth more than a frictionless market.

A final warning about data limitations: the data set only includes trading activities conducted on centralized exchanges (CEXs) and does not include transactions conducted on decentralized exchanges (DEXs). Sanctions are supposed to be more effective on CEXs where the platform acts as a custodian for trader's asset. However, trading on decentralized exchanges accounts for a non-negligible volume of 3 USD billions (see Table A. 4 in the Appendix), a volume that is expected to grow at the expense of centralized exchanges as the centralization risk for DeFi protocols is zero, whereas centralized exchanges and custodian service providers present flaws similar to traditional finance in that feature (FT, 2022).

In terms of future directions for research, the perspectives of cryptocurrencies as payment method are intertwined with those of Central Bank Digital Currency (CBDC). CBDCs would probably transform the current system of payment (Barontini & Holden, 2019), with a reduction in costs and risks for the financial system and a boost in financial inclusion for the individuals (Mancini-Griffoli et al., 2018).[14] The idea of CBDC certainly has potential: a legal tender programmable money, as CBDC would be, could influence far beyond the system of payments. In a scenario with CBDC, we can assist to a different way to conduct the monetary policy agenda, as well as fiscal policy and growth stimulus for national governments.[15]

In conclusion, as it often happens with groundbreaking innovations, within one thing we convey different features that will probably provide their own path for innovation. With cryptocurrencies we have distributed ledger technology (DLT), tokenization of assets, programmability of money, decentralization of finance (DeFi). Each of these topics will surely provide change to the current economic paradigm.

Data availability

Data will be made available on request.

[14] The industry is already taking into account the potential of CBDC. Ripple, for example, is promoting its blockchain network as a suitable CBDC platform to manage a digital currency issued by national and supernational central banks. https://ripple.com/solutions/central-bank-digital-currency/.

[15] The Research in International Business and Finance (RIBS) published in July 2022 a special issue on "The impacts of central bank digital currencies" (edited by Muhammad Ali Nasir, Ahmed Elsayed, John W. Goodell), providing many insightful viewpoints about CBDCs. The list of articles included in the special issue is available at: https://www.sciencedirect.com/journal/research-in-international-business-and-finance/special-issue/10G8K1JZC8P.

APPENDIX

Table A. 1
List of 93 events

Date	Event description	Category
16/07/2013	Vladimir Putin visits the country's biggest military drills since Soviet times, involving around 160,000 troops	Military drills/fight
18/07/2013	Alexei Navalny sentenced to five years in prison	Civil unrest
29/12/2013	Volgograd train station bombing with 16 people dead	Military/non-military incident
01/02/2014	EU-Russia talks on Ukraine & US Dplomatic call	Diplomatic talk/declaration
26/02/2014	Russian invasion of Crimea region	Military drills/fight
02/03/2014	US introduces sanction on Russia	Sanction
03/03/2014	US suspends trade and investment talks with Russia as well as military-to-military cooperation.	Sanction
13/03/2014	Russia starts military exercise near Ukraine	Military drills/fight
16/03/2014	EU & US impose visa restrictions and asset freezes on 21 Russian and Ukrainian/Crimean officials.	Sanction
17/03/2014	Russian annexation of Crimea region	Military drills/fight
20/03/2014	Russia bars 9 US officials from entering Russian territory in response to US sanctions on itself and Crimea. Those sanctioned include former House Speaker John Boehner, Senate Majority Leader Harry Reid, and Senator John McCain.	Sanction
11/04/2014	7 Crimean officials and the Crimean Chernomorneftegaz oil company are added to the US Treasury sanctions list.	Sanction
28/04/2014	Sanctions imposed on 7 individuals and 17 companies linked to President Vladimir Putin. Restrictions placed on Russia's import of U.S. goods deemed to contribute to its military capabilities.	Sanction
12/05/2014	13 individuals as well as Chernomorneftegaz and Feodosia, 2 Crimean energy companies that were expropriated by the de facto authorities on the peninsula, are added to the EU sanctions list. This brings the total number of Russian and Ukrainian nationals targeted with EU asset freezes and visa bans to 61.	Sanction
20/06/2014	US Treasury imposes sanctions on 7 separatists, including military commander Igor Strelkov.	Sanction
12/07/2014	EU imposes sanctions on 11 more individuals.	Sanction
15/07/2014	US Treasury imposes sanctions on 2 major banks and energy companies, 8 arms companies including the Kalashnikov concern, and several officials and separatists.	Sanction
17/07/2014	Malaysia Airlines Flight 17 shots down by Russian controlled forces while flying over Ukraine	Military/non-military incident
23/07/2014	Additional sanctions on Russian individuals and companies	Sanction
26/07/2014	EU adds 15 individuals and 18 entities to sanctions list, bringing the total to 87 individuals and 20 organizations. Individuals added in this round include Chechen leader Ramzan Kadyrov and the director of Russia's FSB security service, Aleksandr Bortnikov.	Sanction
29/07/2014	EU restricts access to capital markets for Russian state-owned banks, imposes an embargo on trade in arms, and restricts exports of dual-use goods and sensitive technologies, especially in the oil sector.	Sanction
04/08/2014	Russia starts military exercise near Ukraine	Military drills/fight
05/08/2014	Heavy fighting erupt in a suburb of the rebel stronghold of Donetsk	Military drills/fight
06/08/2014	US places restrictions on the export of various oil and gas technologies to Russia	Sanction
05/09/2014	Russia and Ukiraine agree to a ceasefire. Russia undergoes financial sanctions	Sanction
11/09/2014	US new sanction on Russia	Sanction
12/09/2014	Financing of Rosneft, Transneft, and Gazpromneft banned. Loans to 5 Russian state banks restricted. Financing of Uralvagonzavod, Oboronprom, and United AIrcraft Corporation (defense manufacturers) banned. 9 Russian defense concerns and 24 individuals added to sanctions list.	Sanction
02/11/2014	Donbas separatist election takes place	Civil unrest
01/12/2014	Russian Central Bank devaluates the Ruble due to oil prices	Central banking intervention
19/12/2014	Export of goods and services to Crimea are banned; imports from Crimea are banned.	Sanction

(*continued on next page*)

Table A. 1 (*continued*)

Date	Event description	Category
24/12/2014	US Sanctions & Ukraine vote for NATO	Sanction
26/12/2014	Russia's new military doctrine is presented	Diplomatic talk/declaration
10/01/2015	Fights at Donetsk airport	Military drills/fight
16/01/2015	US & EU Sanctions are confirmed	Sanction
06/02/2015	NATO expands operations near Russia	Military drills/fight
11/02/2015	Minks talks start	Diplomatic talk/declaration
12/02/2015	Minsk agreement and end of Donbas war between Russian separatist groups and Armed forces of Ukraine	Military drills/fight
16/02/2015	19 individuals and 9 entities added to the EU sanctions list, including the Sparta, Somali, Zarya, Oplot, Kalmius, and Smert' separatist battalions	Sanction
26/02/2015	Russia annexation of 2 breakaway Georgian regions	Diplomatic talk/declaration
02/03/2015	US extends economic sanctions by one year.	Sanction
04/03/2015	All US sanctions imposed on Russia in 2014 extended by one year.	Sanction
11/03/2015	US imposes sanctions on 14 individuals, the Russian National Commercial Bank, and the Eurasian Union of Youth.	Sanction
14/03/2015	EU sanctions against 151 individuals and 37 entities extended until September 15, 2015.	Sanction
22/06/2015	EU economic sanctions against Russia extended to January 31, 2016.	Sanction
24/06/2015	Law allowing punishments for foreign banks doing business with sanctioned Russian individuals and entities comes into force.	Sanction
30/06/2015	US Treasury sanctions 11 individuals and 15 entities.	Sanction
07/08/2015	US imposes sanctions on the Yuzhno-Kirinskoye oil field.	Sanction
14/09/2015	EU visa bans and asset freezes against 149 individuals and 37 entities are extended until March 15, 2016.	Sanction
29/09/2015	Russia begins airstrikes in Syria	Military drills/fight
21/12/2015	EU economic sanctions against Russia extended to July 31, 2016	Sanction
22/12/2015	US Treasury targets 34 individuals and entities for sanctions.	Sanction
10/03/2016	EU sanctions against 146 individuals and 37 entities extended to September 15, 2016.	Sanction
17/06/2016	EU sanctions on Crimea extended to June 23, 2017.	Sanction
01/09/2016	US designates 17 individuals and a number of entities for sanctions, mostly local subsidiaries of Gazprom.	Sanction
15/09/2016	EU sanctions against 146 individuals and 37 entities extended to March 2017.	Sanction
28/09/2016	Confirmation that a missile from Russia is blamed for Malaysia Airlines Flight 17 tragedy	Military/non-military incident
09/11/2016	EU imposes visa bans and asset freezes on 6 Duma lawmakers from Crimea.	Sanction
15/11/2016	US imposes sanctions on six Duma lawmakers from Crimea.	Sanction
19/12/2016	EU extends economic sanctions to July 31, 2017.	Sanction
20/12/2016	US imposes visa bans and asset freezes on seven individuals and a number of companies involved in construction and logistics in Crimea.	Sanction
23/12/2016	US designates 23 Russian companies for sanctions.	Sanction
13/01/2017	US economic sanctions extended by one year.	Sanction
15/03/2017	EU sanctions against 150 individuals and 37 entities extended by six months.	Sanction
19/06/2017	EU sanctions on Crimea extended to June 23, 2018.	Sanction
20/06/2017	US imposes sanctions on 38 individuals and entities, including the military company PMC Wagner.	Sanction

(*continued on next page*)

Table A. 1 (*continued*)

Date	Event description	Category
29/06/2017	Economic sanctions extended to January 31, 2018.	Sanction
04/08/2017	EU designates three Russian citizens and three companies for sanctions in connection with the delivery of Siemens gas turbines to Crimea.	Sanction
14/09/2017	Sanctions against 149 individuals and 38 entities extended to March 15, 2018.	Sanction
21/11/2017	The EU imposes sanctions on Dmitry Ovsyannikov, the Russia-installed governor of Sevastopol.	Sanction
22/12/2017	EU extends economic sanctions on Russia to July 31, 2018.	Sanction
26/01/2018	US imposes sanctions on 21 individuals and nine companies.	Sanction
30/01/2018	US publishes the "Kremlin dossier," a list of 210 people close to Vladimir Putin who might be designated for sanctions. The list includes all members of the Russian government, members of the presidential administration, heads of state firms, "siloviki" (former and current state-security officers), and oligarchs.	Sanction
02/03/2018	US economic sanctions extended by one year.	Sanction
12/03/2018	EU sanctions against 150 individuals and 38 organizations extended to 15 September 2018	Sanction
06/04/2018	US designates 38 Russian businessmen for visa bans and asset freezes to punish Russian "malign activity" worldwide.	Sanction
17/06/2018	EU sanctions on Crimea extended to June 23, 2019.	Sanction
09/07/2018	EU economic sanctions on Russia extended to January 31, 2019.	Sanction
21/01/2019	Additional sanctions on Russian individuals and companies	Sanction
21/04/2021	Pro-Navalny prostest & Navalny critical condition	Civil unrest
26/04/2021	Russian naval military drills in Black Sea	Military drills/fight
19/05/2021	Russia advances unlawful maritime claim in the Arctic	Diplomatic talk/declaration
05/08/2021	Russia and Belarus to press ahead with military drills criticised by Ukraine	Military drills/fight
10/08/2021	Russia and China joint military drills	Military drills/fight
07/09/2021	Russia and Belarus hold huge military drills	Military drills/fight
13/09/2021	Russia and Belarus continue joint massive military drills	Military drills/fight
20/09/2021	Russia military drills in Black Sea & Ukraine and US military drills nearby	Military drills/fight
16/11/2021	Russia performs 'reckless and dangerous' anti-satellite missile test	Military/non-military incident
04/12/2021	White House: Russia ready to invade Ukraine	Diplomatic talk/declaration
20/01/2022	White House: Russia can invade Ukraine	Diplomatic talk/declaration
21/01/2022	US military assistance to Ukraine	Military drills/fight
24/01/2022	NATO sends reinforcements and U.S. puts troops on alert as Ukraine tensions rise	Military drills/fight
21/02/2022	Russian President Vladimir Putin announced that Russia recognizes the independence of two pro-Russian breakaway regions in eastern Ukraine. This leads to a first round of economic sanctions from NATO countries the following day.	Diplomatic talk/declaration
24/02/2022	Special military operation starts in east Ukraine	Military drills/fight

Table A. 2
Panel regression analysis with blockchain controls.

Variable	(1)	(2)	(3)	(4)
	ΔVolRUB	ΔVolRUBbps	Ln(VolRUB)	Ln(VolRUBbps)
RUB	1.468**	1.142*	2.426	2.426
	(14.398)	(7.041)	(2.093)	(2.093)
Crypto	0.335*	0.561	0.337	0.337
	(10.707)	(3.257)	(1.753)	(1.753)
Sp500	−1.655	−0.896	1.169	1.169
	(-4.104)	(-1.256)	(2.220)	(2.220)

(*continued on next page*)

Table A. 2 (*continued*)

Variable	(1)	(2)	(3)	(4)
	ΔVolRUB	ΔVolRUBbps	Ln(VolRUB)	Ln(VolRUBbps)
ΔVolume	0.465			
	(1.855)			
Event	0.265***	0.203*	0.348*	0.348*
	(511.816)	(10.090)	(11.216)	(11.216)
Ln(Volume)			0.431	−0.569
			(1.911)	(-2.519)
Constant	−0.012	0.097	−2.299	6.912
	(-0.201)	(1.420)	(-0.541)	(1.627)
Observations	3846	3846	3846	3846
R-squared	0.176	0.021	0.633	0.505
Std. Errors	Robust	Robust	Robust	Robust
Year FE	YES	YES	YES	YES
Control	Blockchain	Blockchain	Blockchain	Blockchain
Sample	BTC, ETH	BTC, ETH	BTC, ETH	BTC, ETH

The Table provides panel regression analysis for model presented in Equation (1) with the addition of Blockchain control variables. Blockchain control variables are available only for Bitcoin and Ethereum networks.
t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

Table A. 3
Time series regression analysis with lagged values.

Model	(1)	(2)	(3)	(4)	(5)
Cryptocurrency	BTC	ETH	DASH	XRP	USDT
AR(1)	0.090***	0.041***	0.024***	0.059***	0.047***
	(7.333)	(2.832)	(3.374)	(4.706)	(4.570)
AR(2)	0.080***	0.049***	0.007	0.084***	0.065***
	(6.677)	(3.503)	(0.968)	(6.181)	(6.239)
AR(3)	0.074***	0.029**	0.012*	0.049***	0.024**
	(6.415)	(2.021)	(1.652)	(3.560)	(2.378)
AR(4)	0.068***	0.035**	−0.000	0.060***	0.027***
	(5.225)	(2.421)	(-0.061)	(4.647)	(2.752)
Event	0.279***	0.499***	0.062	0.453***	0.235***
	(4.004)	(3.858)	(0.653)	(4.841)	(2.604)
Event.L1	0.191**	0.346***	−0.025	0.354***	0.095
	(2.395)	(2.674)	(-0.299)	(3.148)	(0.813)
Event.L2	0.002	0.232	−0.002	0.269**	0.006
	(0.037)	(1.594)	(-0.018)	(2.297)	(0.063)
Event.L3	0.010	0.039	−0.023	0.280***	−0.006
	(0.132)	(0.355)	(-0.243)	(2.584)	(-0.076)
Event.L4	−0.101	0.150	−0.044	0.332***	0.112
	(-0.887)	(1.411)	(-0.445)	(3.239)	(1.145)
Constant	0.190	0.158	2.464***	−0.715	4.653***
	(0.532)	(0.322)	(9.637)	(-1.307)	(10.035)
Observations	2290	1698	1386	1305	1324
R-squared	0.855	0.460	0.393	0.610	0.955
Std. Errors	Robust	Robust	Robust	Robust	Robust
Year dummy	YES	YES	YES	YES	YES
Control	YES	YES	YES	YES	YES

The Table provides time series regression analysis for model presented in Equation (1) and for each cryptocurrency with lagged values up to one week for all considered variables. The dependent variable is LnVolRUB, the control variables are RUB, Crypto, Ln(Volume) and Sp500.
t-values are in parentheses.
***p < .01, **p < .05, *p < .1.

Table A. 4
Ranking for decentralized and centralized exchanges as of mid-2022

Ranking	DEX		CEX	
	Name	24 h Volume	Name	24 h Volume
1	Uniswap (V3)	874.37	Binance	11,005.12
2	dYdX	846.35	Deepcoin	4698.46
3	Kine Protocol	245.29	CITEX	4106.73
4	PancakeSwap (V2)	230.47	TOKENCAN	3299.89

(*continued on next page*)

Table A. 4 (*continued*)

Ranking	DEX		CEX	
	Name	24 h Volume	Name	24 h Volume
5	Honeyswap	141.84	Biconomy Exchange	2425.70
6	ApolloX DEX	105.34	P2PB2B	2225.58
7	Uniswap (V2)	77.07	AAX	2003.37
8	DODO (Ethereum)	59.45	HitBTC	1845.49
9	Curve Finance	46.64	Pionex	1816.28
10	Jupiter	44.11	FMFW.io	1809.69
11	Perpetual Protocol	40.49	OKX	1803.31
12	SushiSwap	31.58	BitCoke	1780.26
13	QuickSwap	30.32	Coinsbit	1684.60
14	DODO (BSC)	29.56	Hotcoin Global	1678.70
15	Biswap	28.48	DigiFinex	1664.15
16	Uniswap (V3) (Potimism)	25.62	Upbit	1592.17
17	iZiSwap	24.81	Huobi Global	1578.92
18	Serum DEX	22.50	CoinW	1543.43
19	Sun.io	21.38	Bitrue	1541.52
20	Kwenta	21.19	BitMart	1490.99
Top 20		2946.88		51,594.36

The Table provides the ranking for decentralized (DEX) and centralized (CEX) exchanges that allow trading for cryptocurrency. Ranking is organized by daily volume in millions of US dollars. The ranking is updated as of July 24, 2022. Source: Coinmarketcap.com.

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